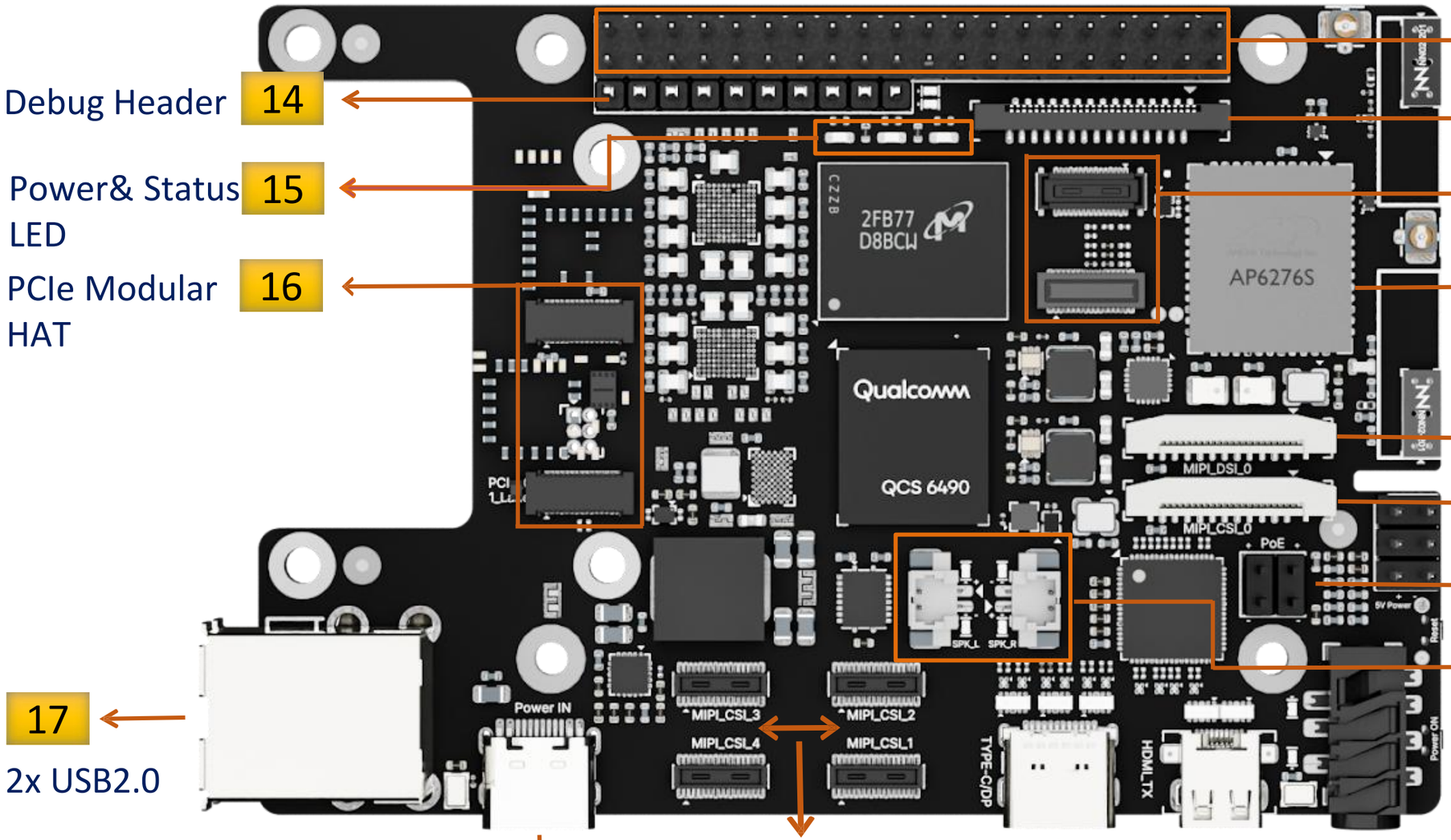
TOP SIDE



Debug Header 14

Power& Status LED 15

PCIe Modular HAT 16

17 2x USB2.0

13 2 X 20 GPIOs

12 30 GPIOs FPC

11 UFS Module HAT

10 WIFI 6 + BT5.2

09 MIPI DSI0 TX

08 MIPI CSI0 RX

07 PoE Connector

06 2x Speaker Out

01 TYPE-C Power 12V/5V

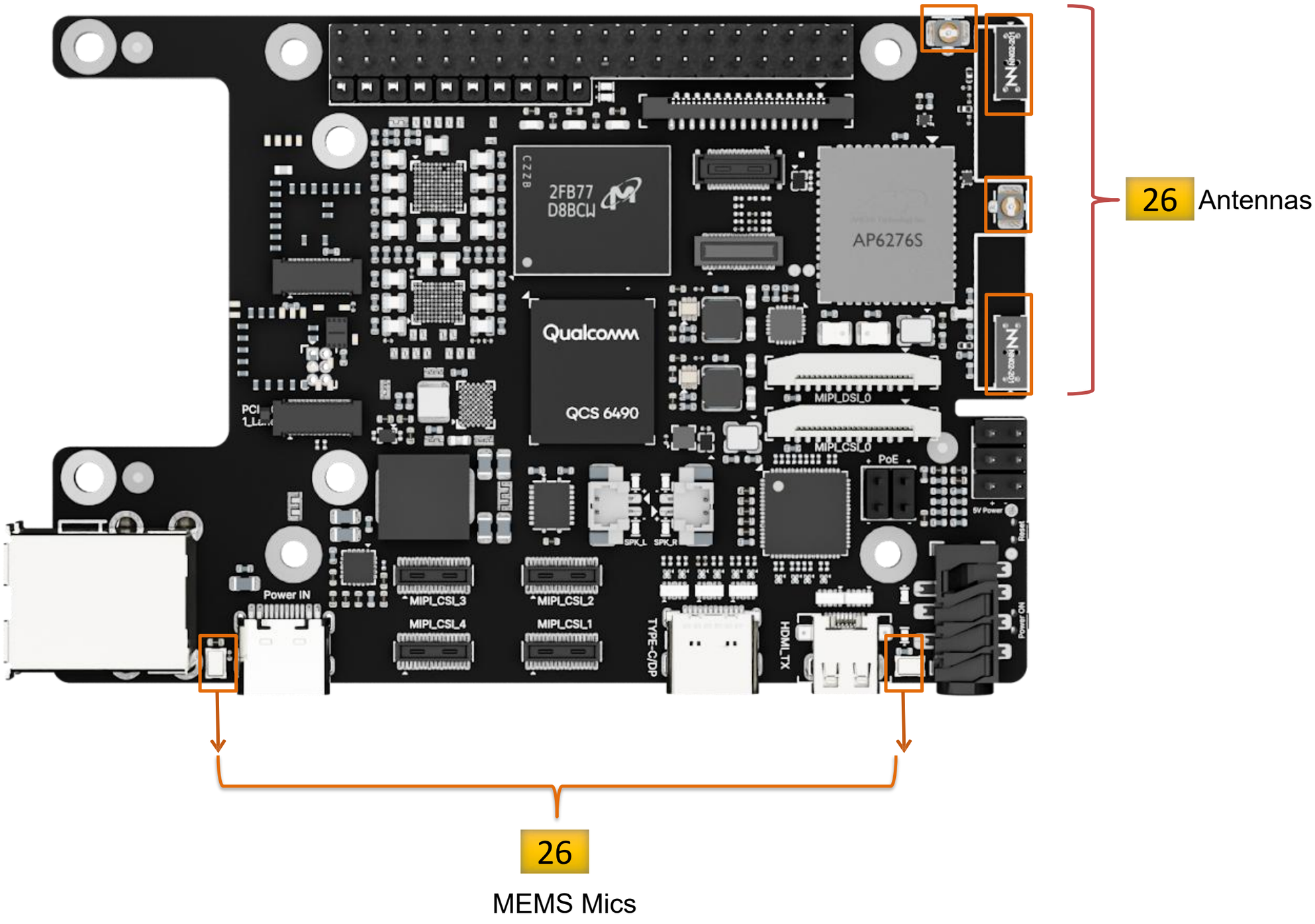
02 MIPI CSI 1-4 RX

03 TYPE-C/DP Port 5V Input

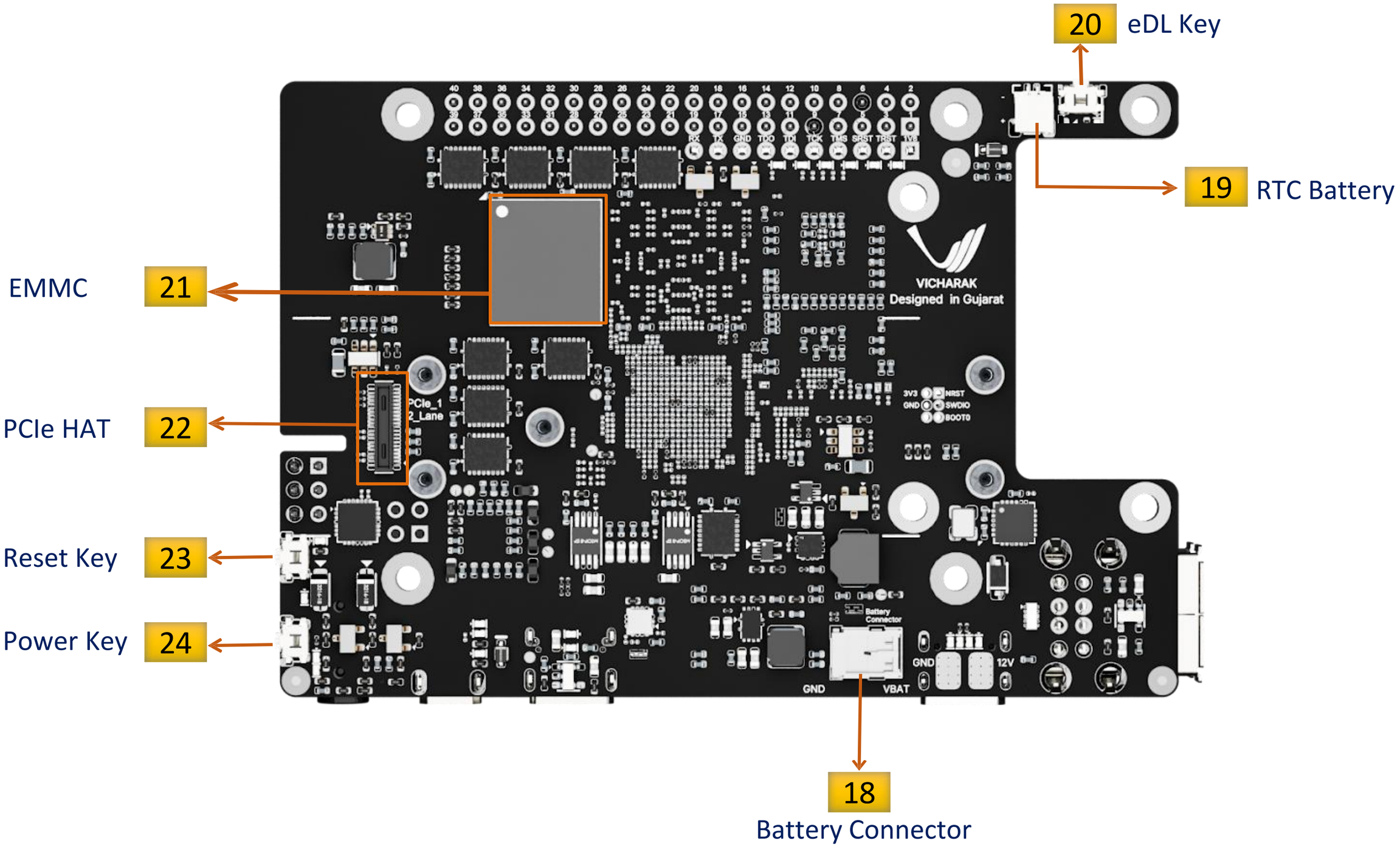
04 HDMI TX

05 Audio Jack 3.5mm

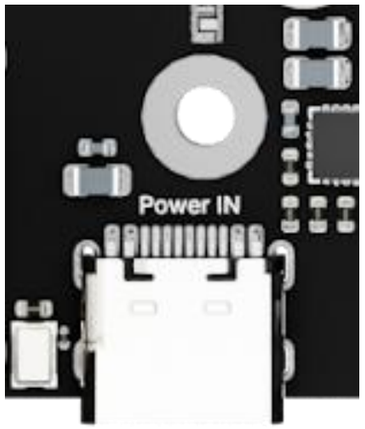
TOP SIDE


26 Antennas
2FB77
D8BCW
AP6276S
Qualcomm
QCS 6490
MIPI_DSI_0
MIPI_CSI_0
PoE
5V Power
Reset
Power ON
SPK_L
SPK_R
MIPI_CSI_3
MIPI_CSI_2
MIPI_CSI_4
MIPI_CSI_1
Power IN
TYPE-C/DP
HDMI_TX
26
MEMS Mics

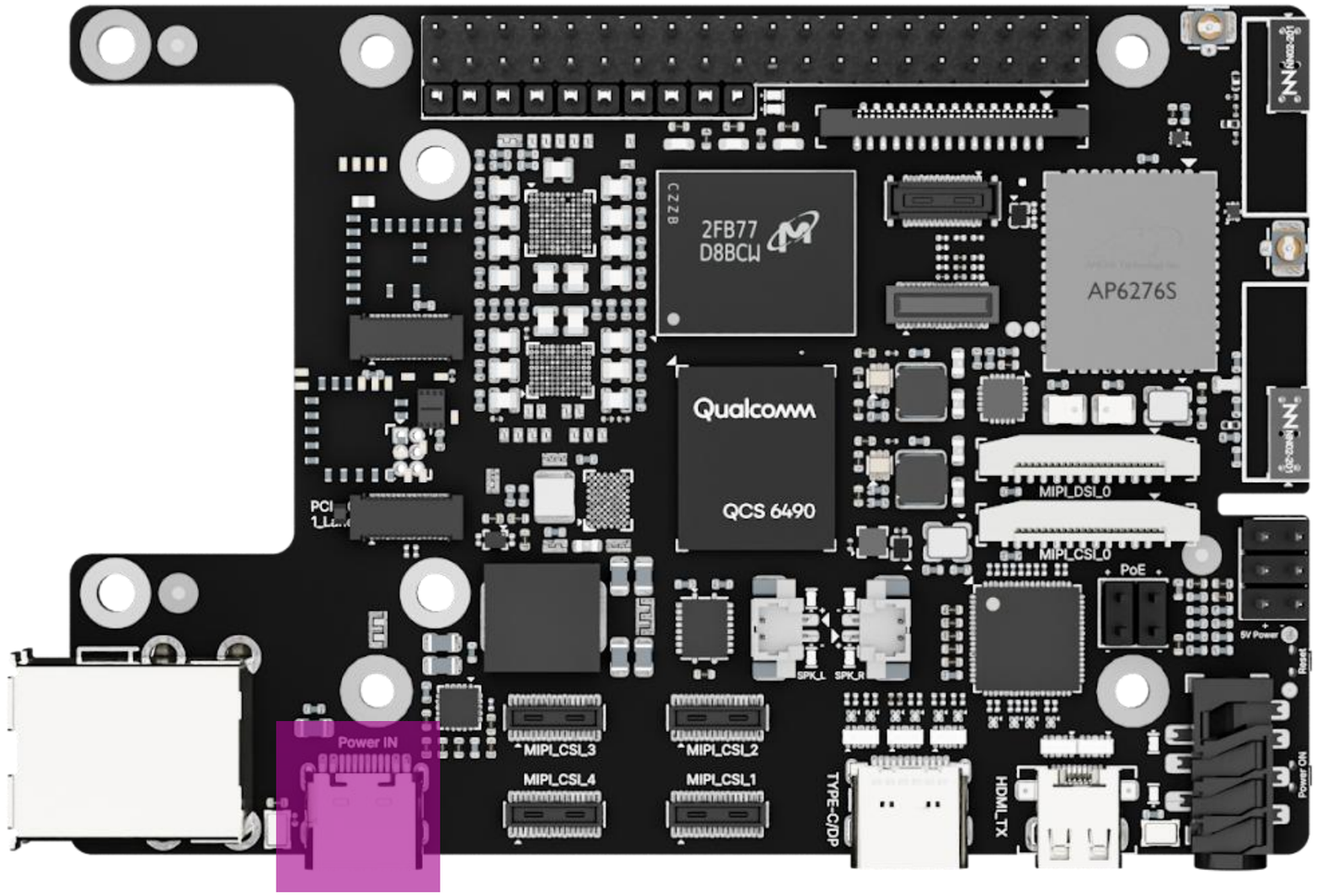
BOTTOM SIDE


20 eDL Key
19 RTC Battery
EMMC
21
PCIe HAT
22
Reset Key
23
Power Key
24
18
Battery Connector
VICHARAK
Designed in Gujarat
PCIe_1
2_Lane
3V3
NRST
GND
SWDIO
BOOT0
Battery
Connector
GND
VBAT
GND
12V

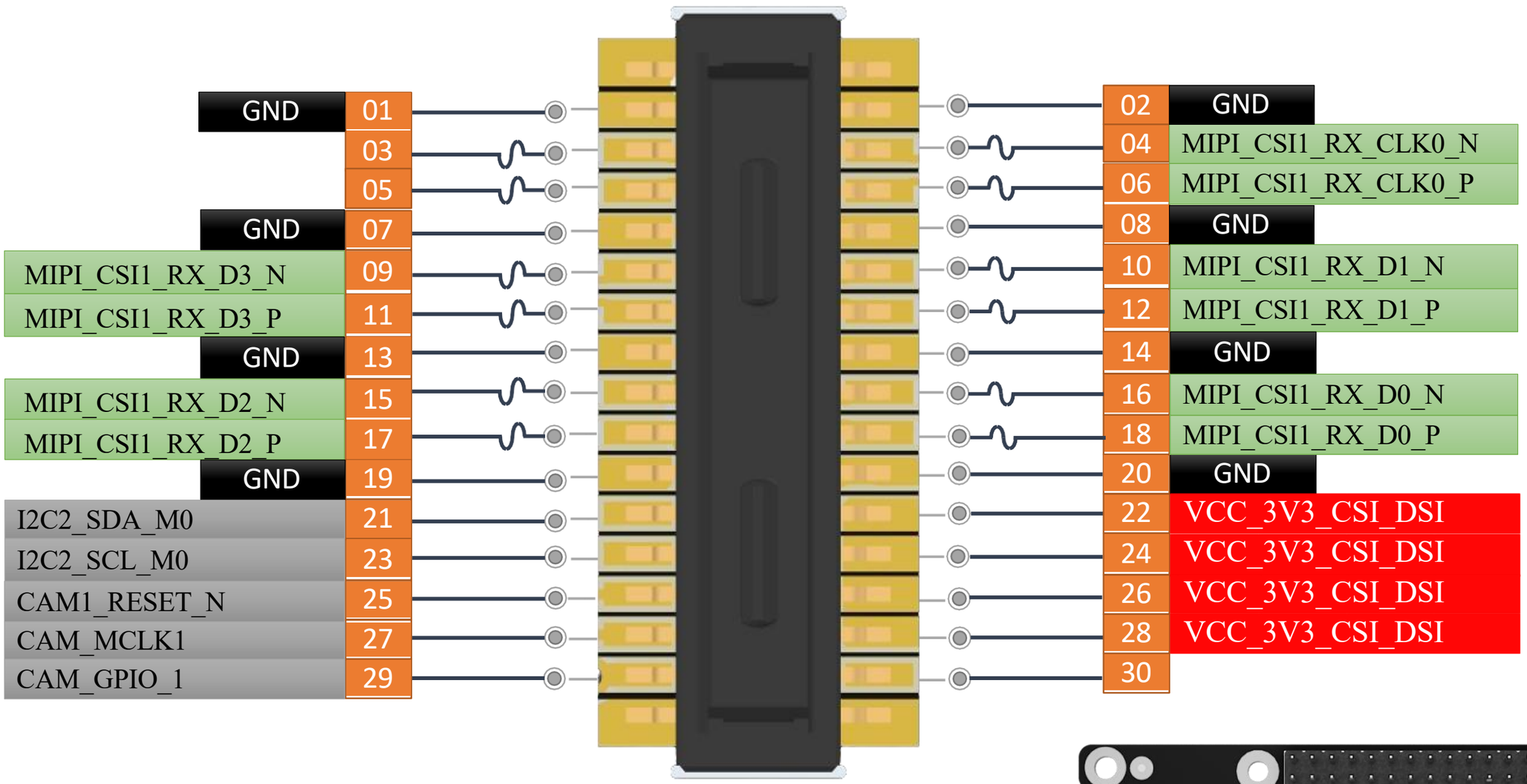
01 Type-C PD (12V Input) / 5V Input



Type-C Power Input
(12V / 5V)



02 MIPI_CSI 1&2 RX



MIPI_CSI 1 to 4 RX

On-Board Connector

30 Pin

0.4mm Pitch

Current Rating (Max): 300mA each pin

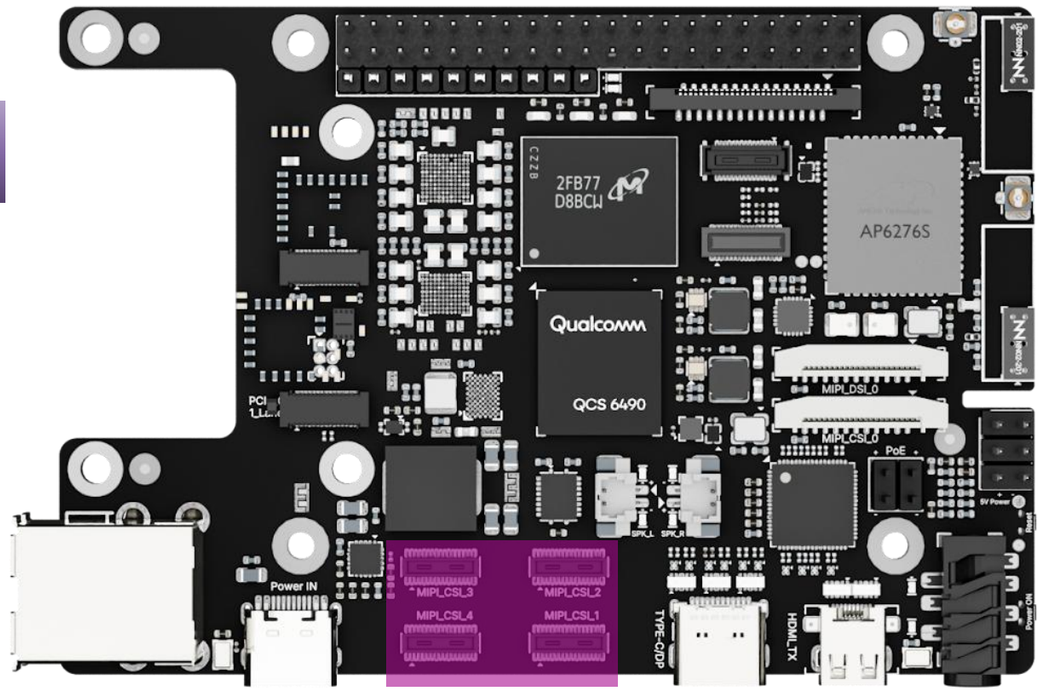
MPN : DF40C(2.0)-30DP-0.4V(51)

Mating Connector

30 Pin

0.4mm Pitch

MPN : DF40C(2.0)-30DS-0.4V(51)

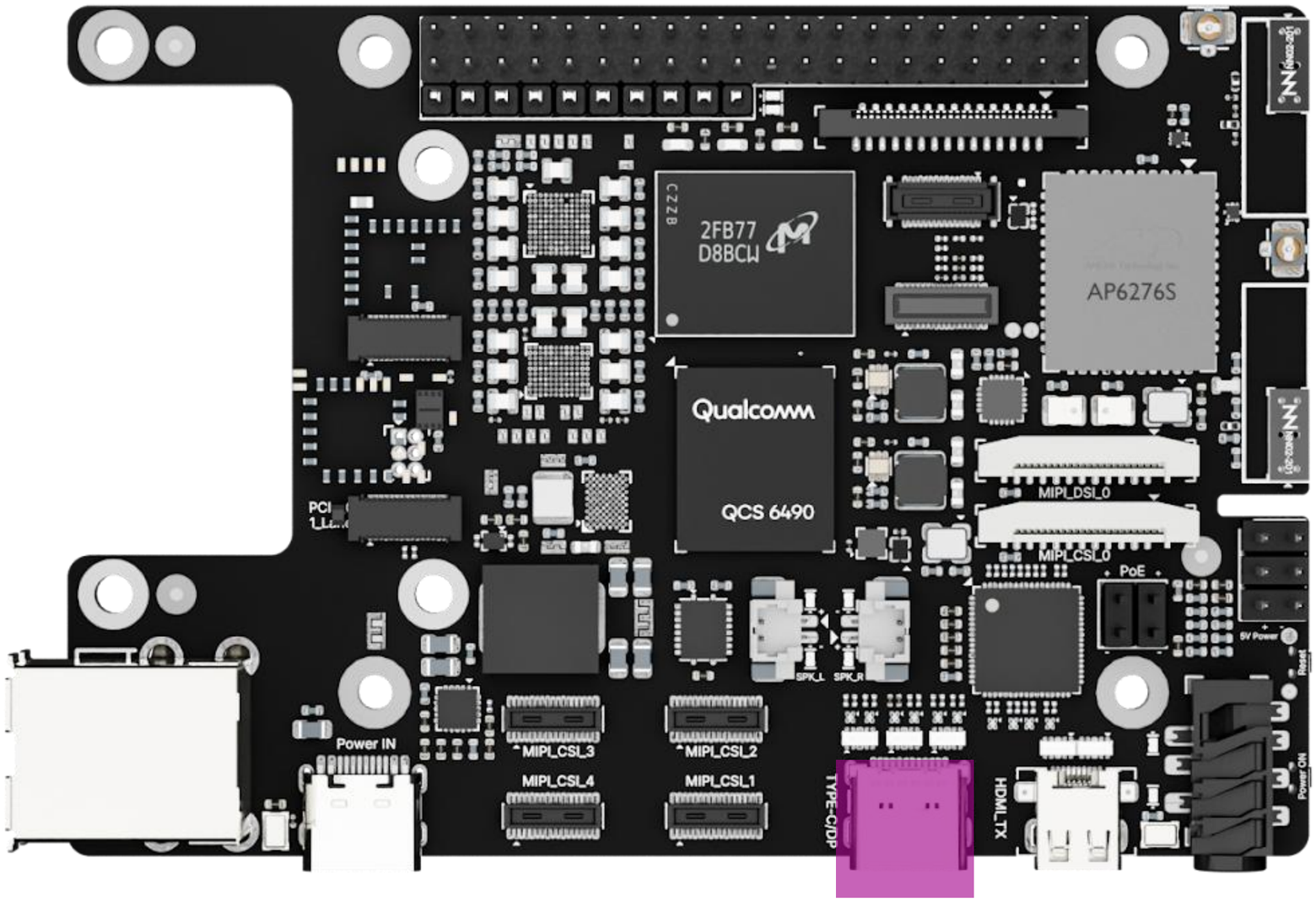


03 Type-C/DP Port

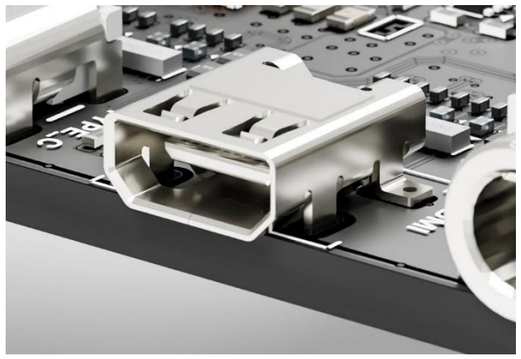


USB 3.0 TYPE-C/DP

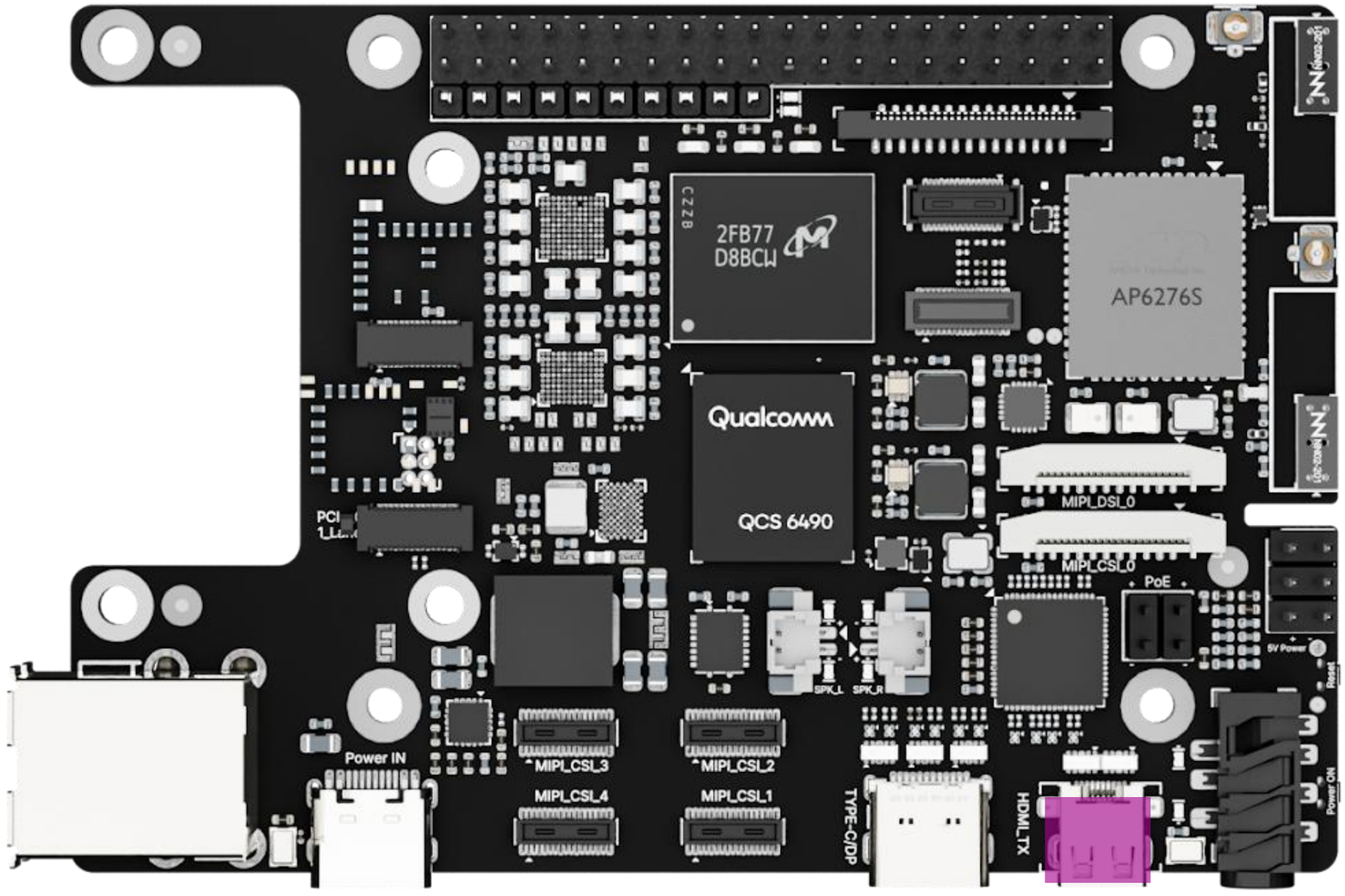
5V Input



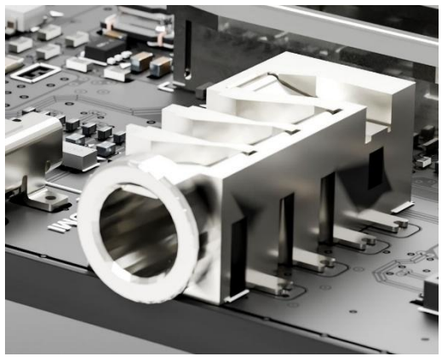
04 HDMI TX Port



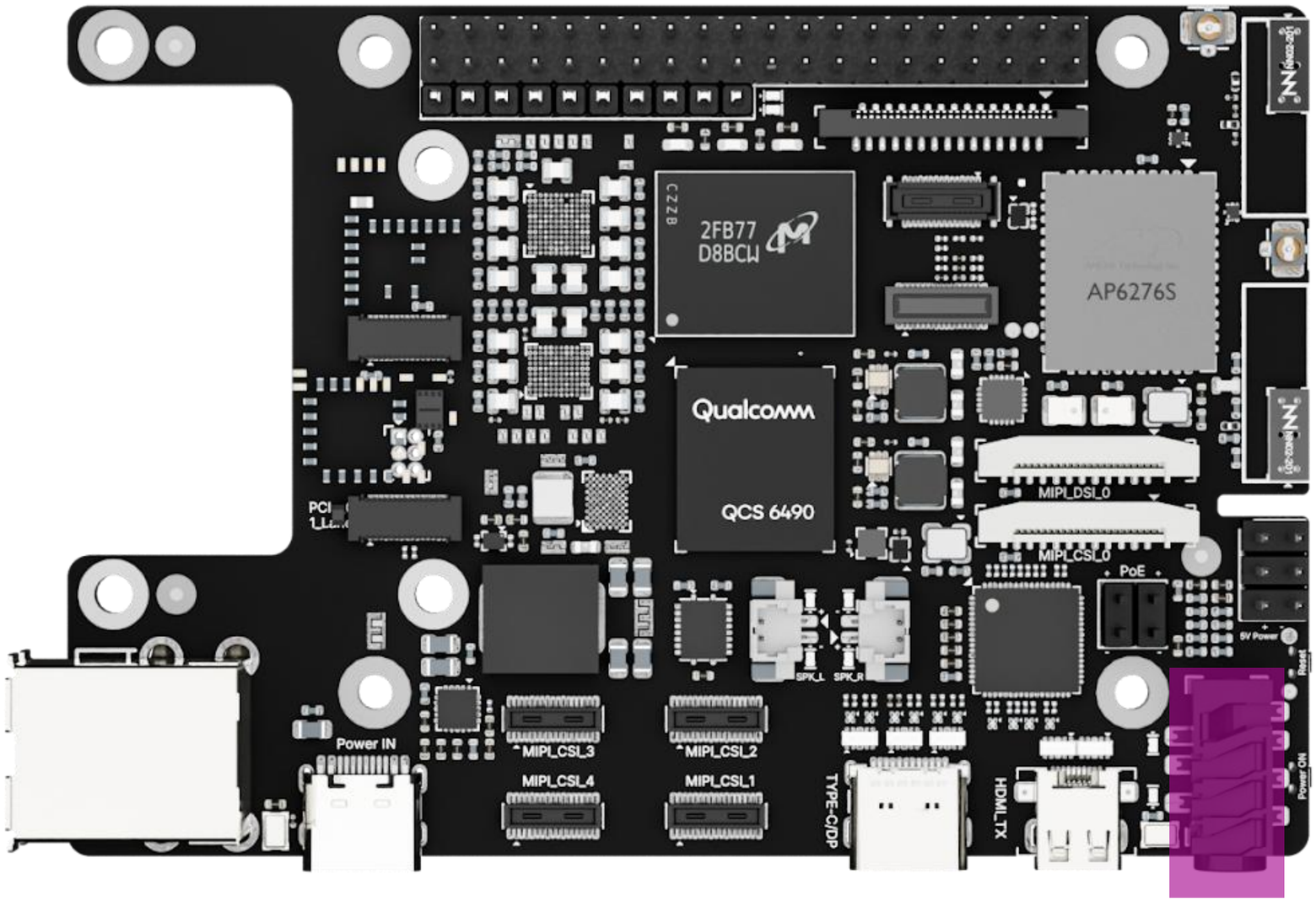
HDMI 2.1 Port



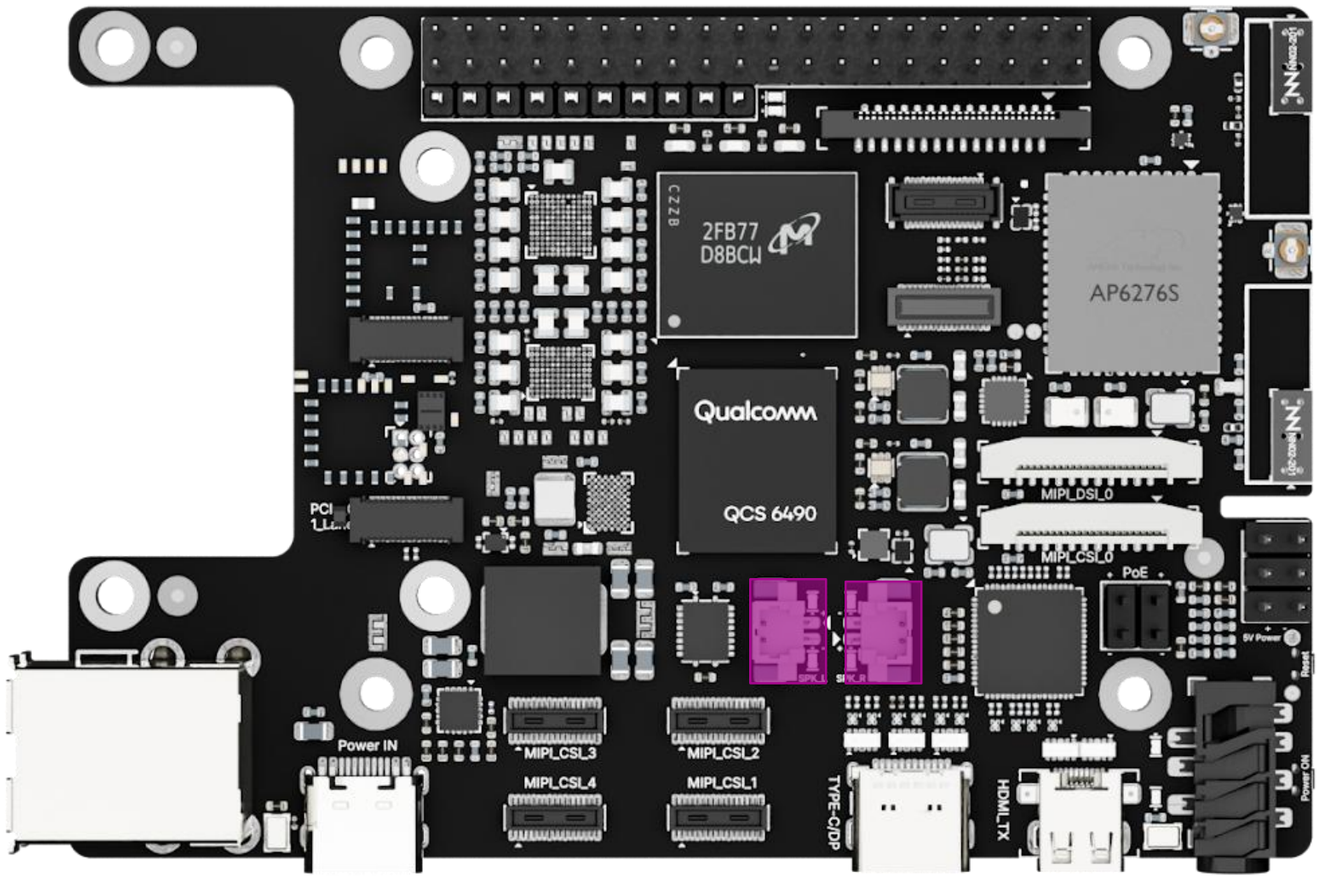
05 Audio Jack 3.5mm



3.5mm Audio Jack



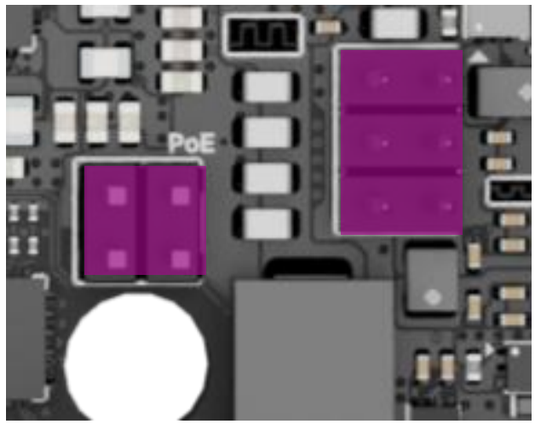
06 Speaker Out



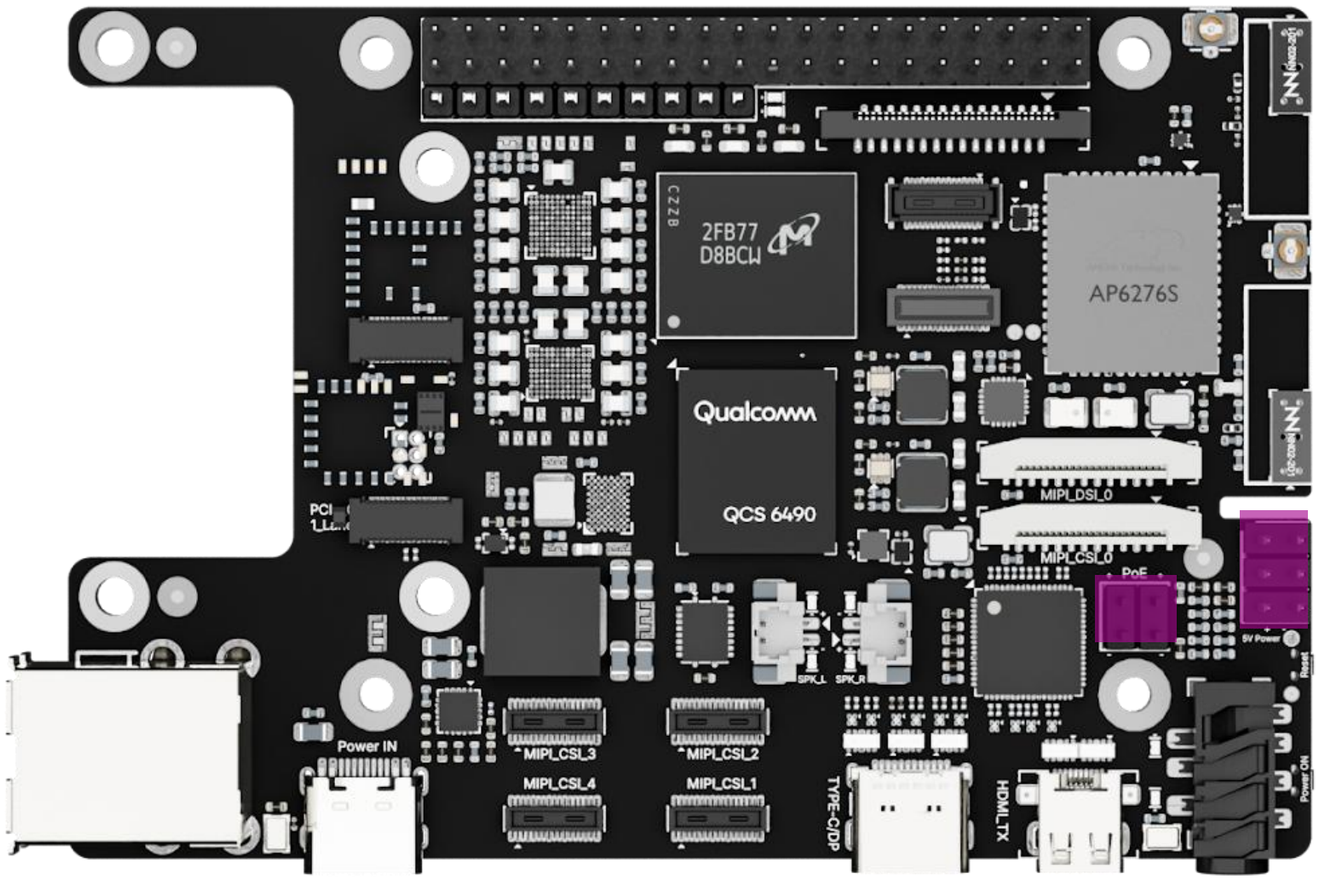
W2B Connector
2Pin
1.25mm Pitch
Current rating (max): 1A each pin
MPN : WAFER-125L-2A

Speaker
load : 4 ohm
Power :3W

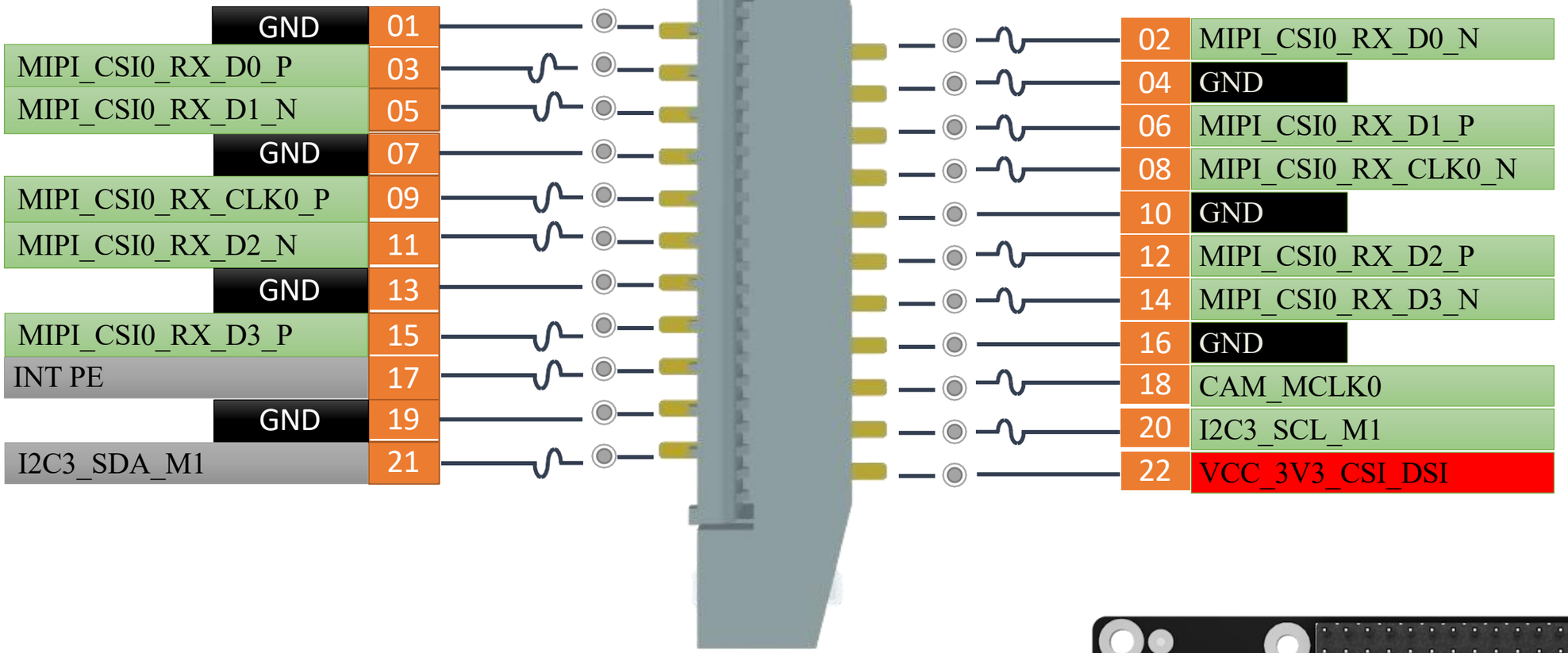
07 PoE Connector



PoE HAT Connector

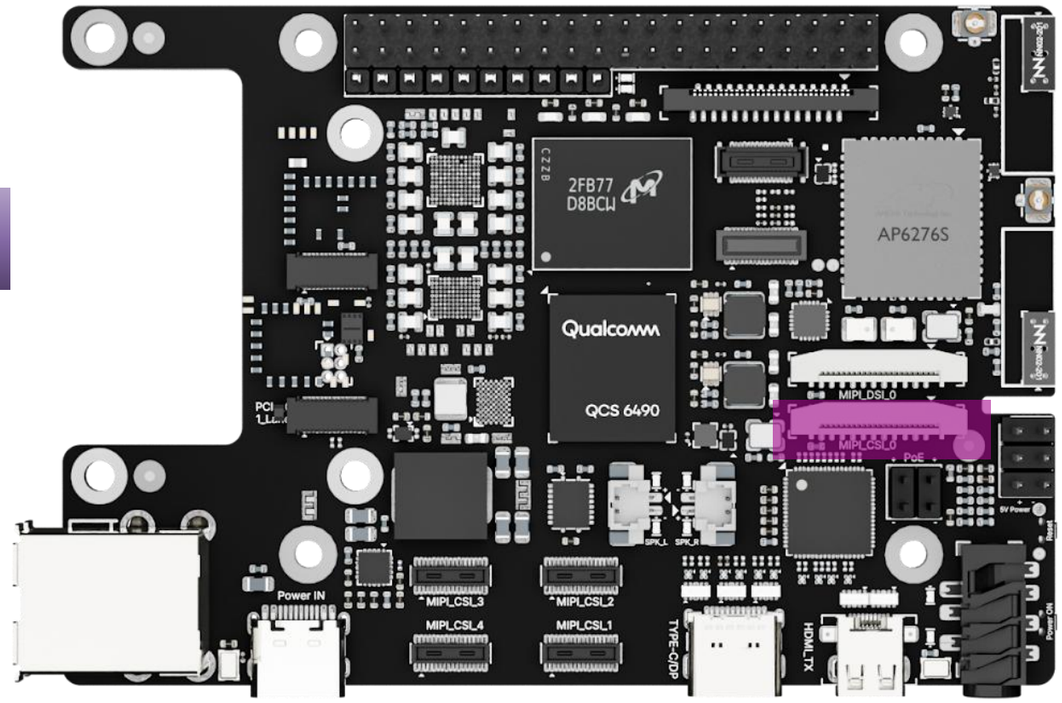


08 MIPI CSI RX PORT0

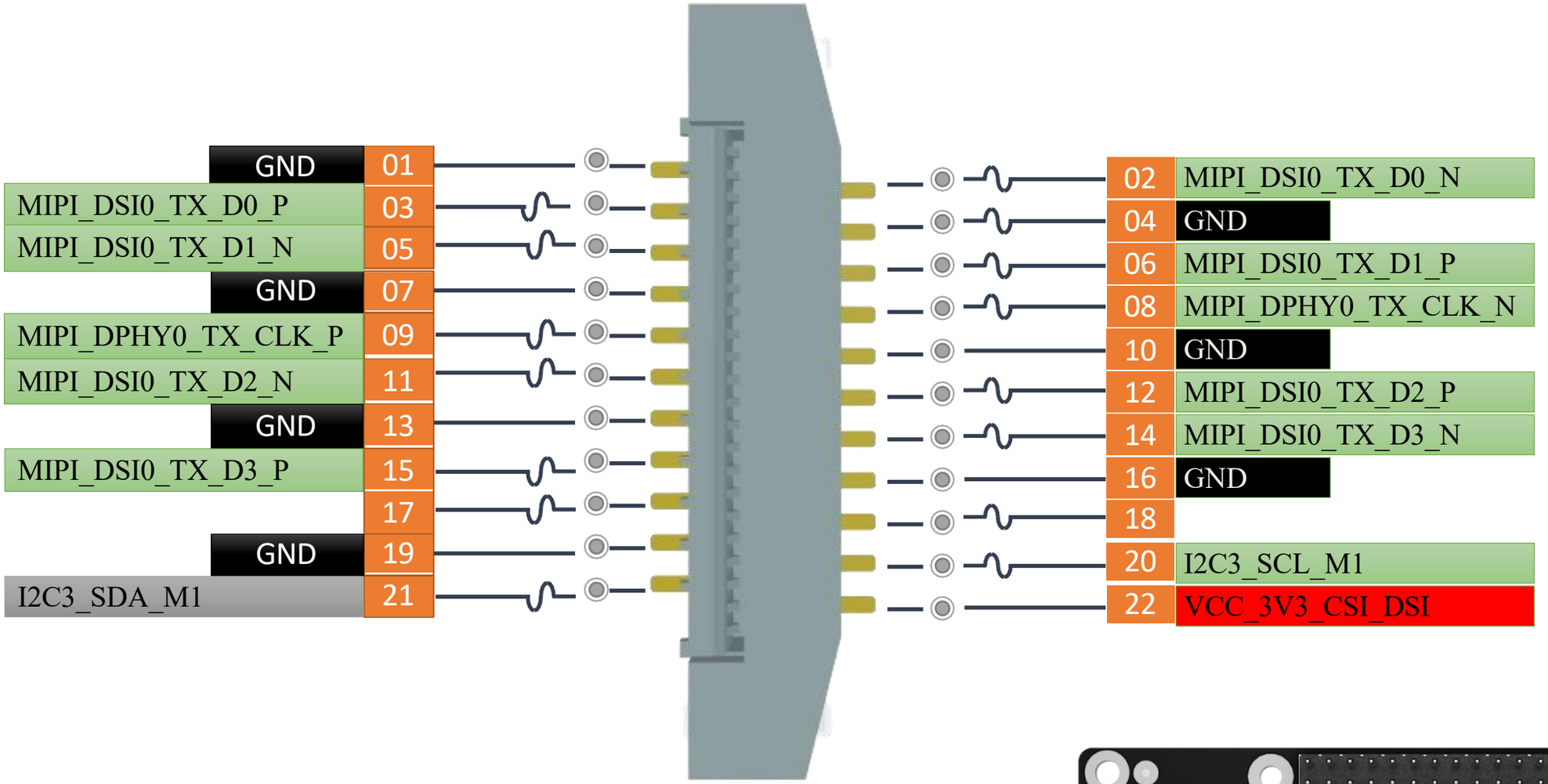


MIPI CSI 0 RX

FPC Connector
RPI Compatible
22 Pin
0.4mm Pitch
Current Rating (Max): 300mA each pin
MPN : AFC11-S22ICA-00

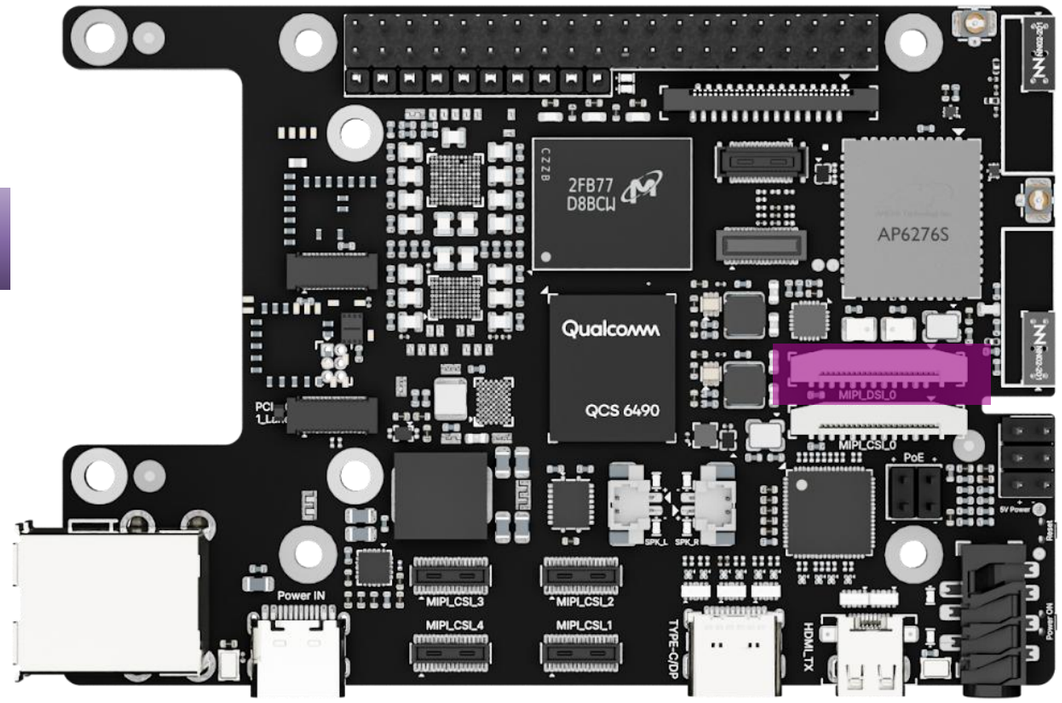


09 MIPI_DSI 0_TX

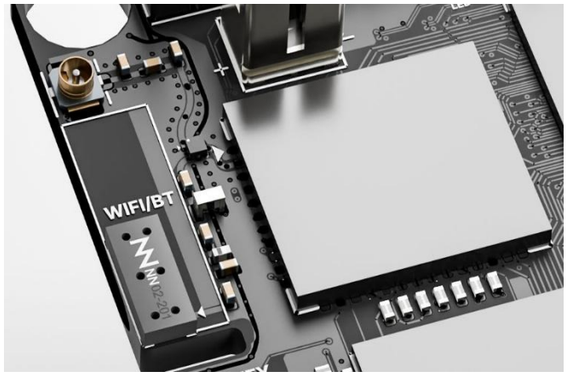


MIPI_DPHY0_TX

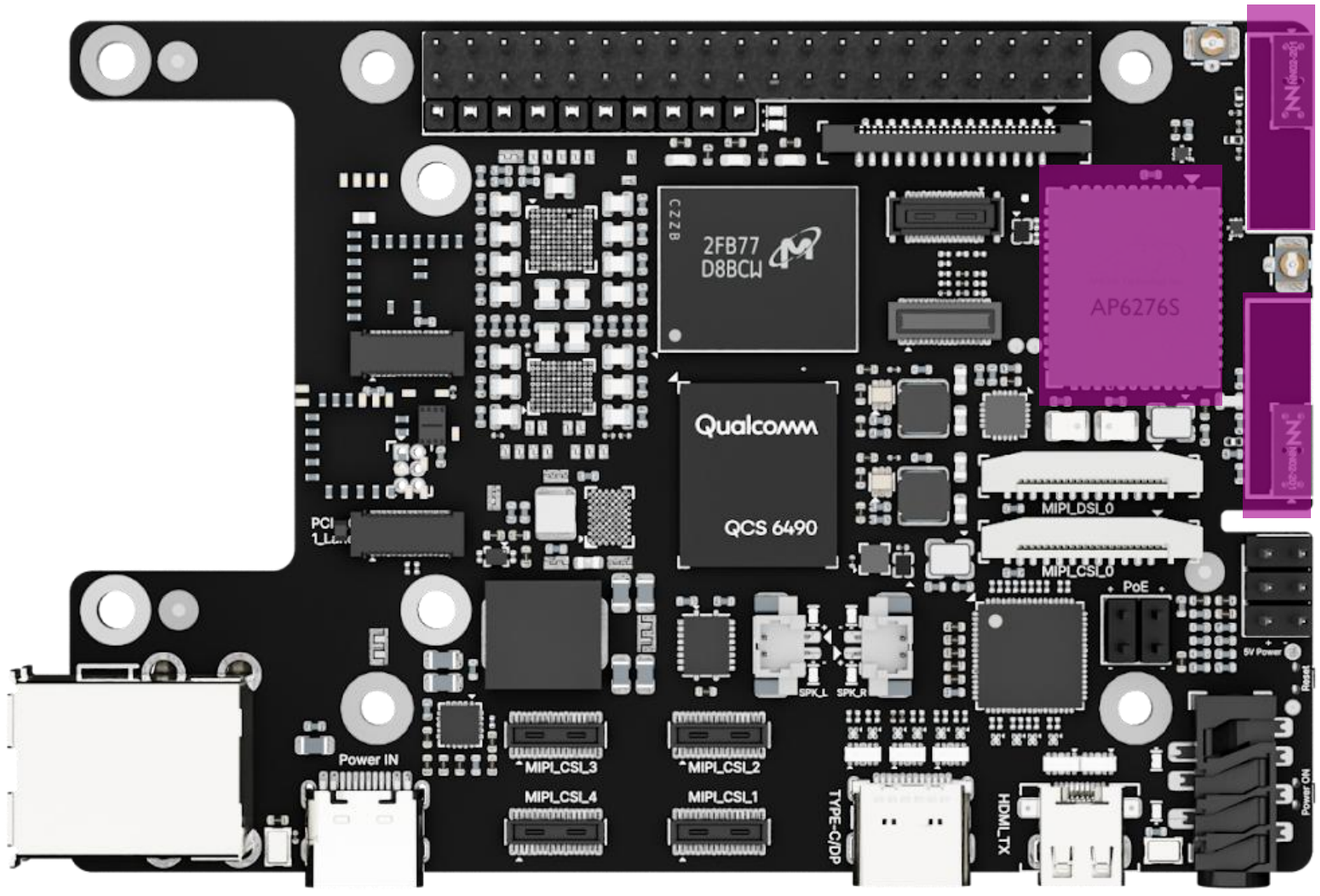
FPC Connector
RPI Compatible
22 Pin
0.4mm Pitch
Current Rating (Max): 300mA each pin
MPN : AFC11-S22ICA-00



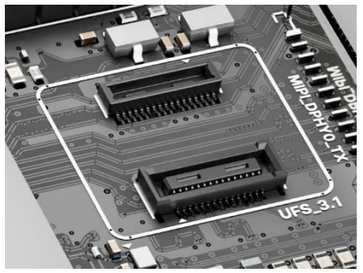
Wi-Fi 6 + BT 5.2 Module



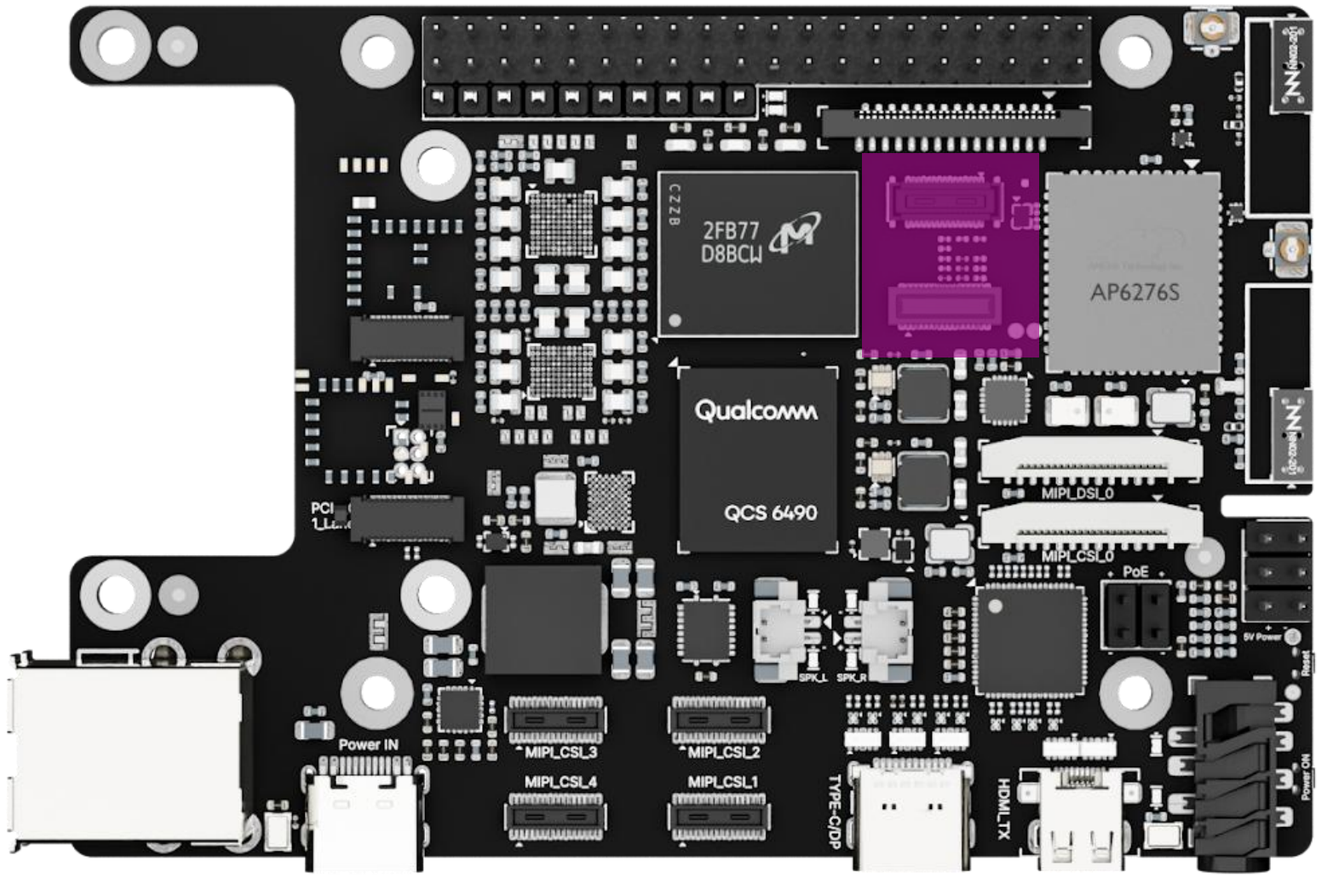
WI-FI 6/BT 5.4
AP6276S



Modular UFS Port



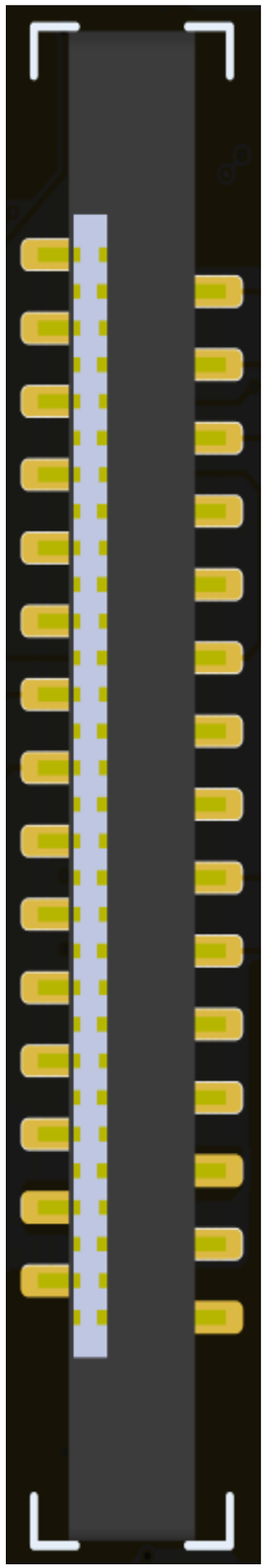
Modular UFS PORT



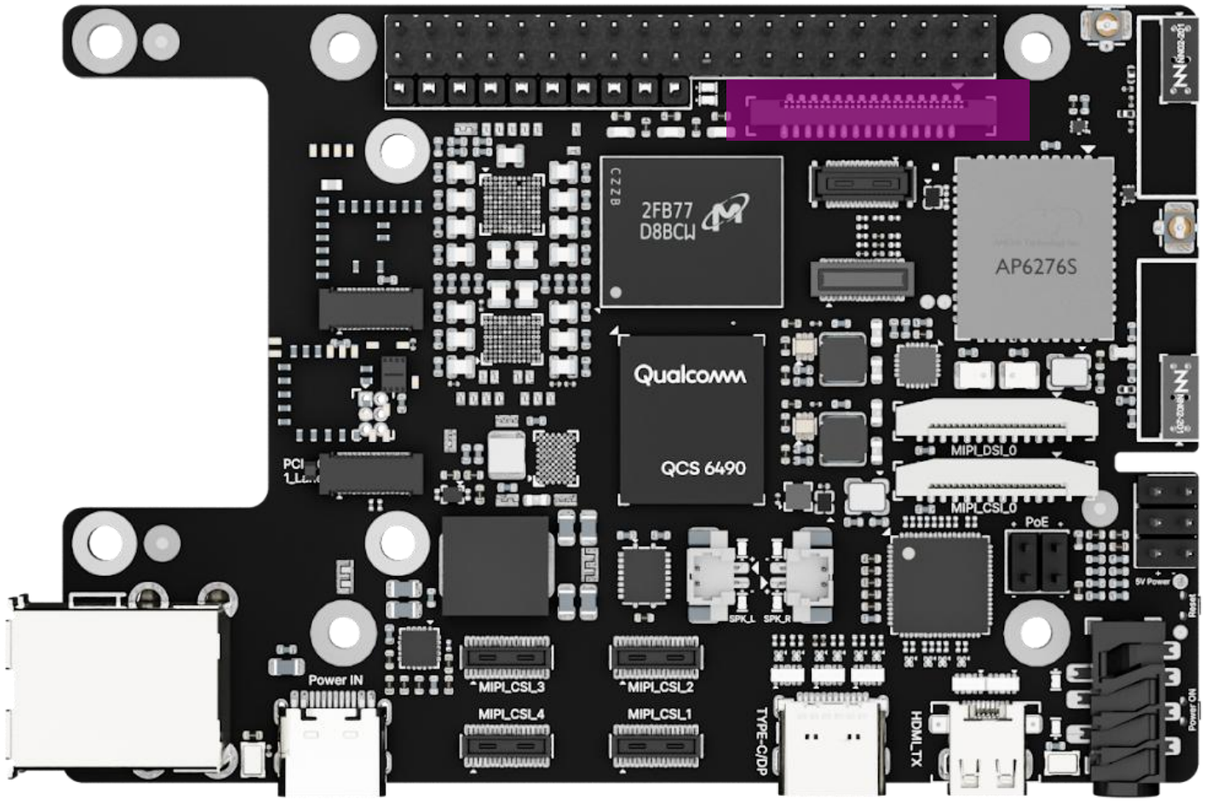
30 Pin GPIO FPC

1st

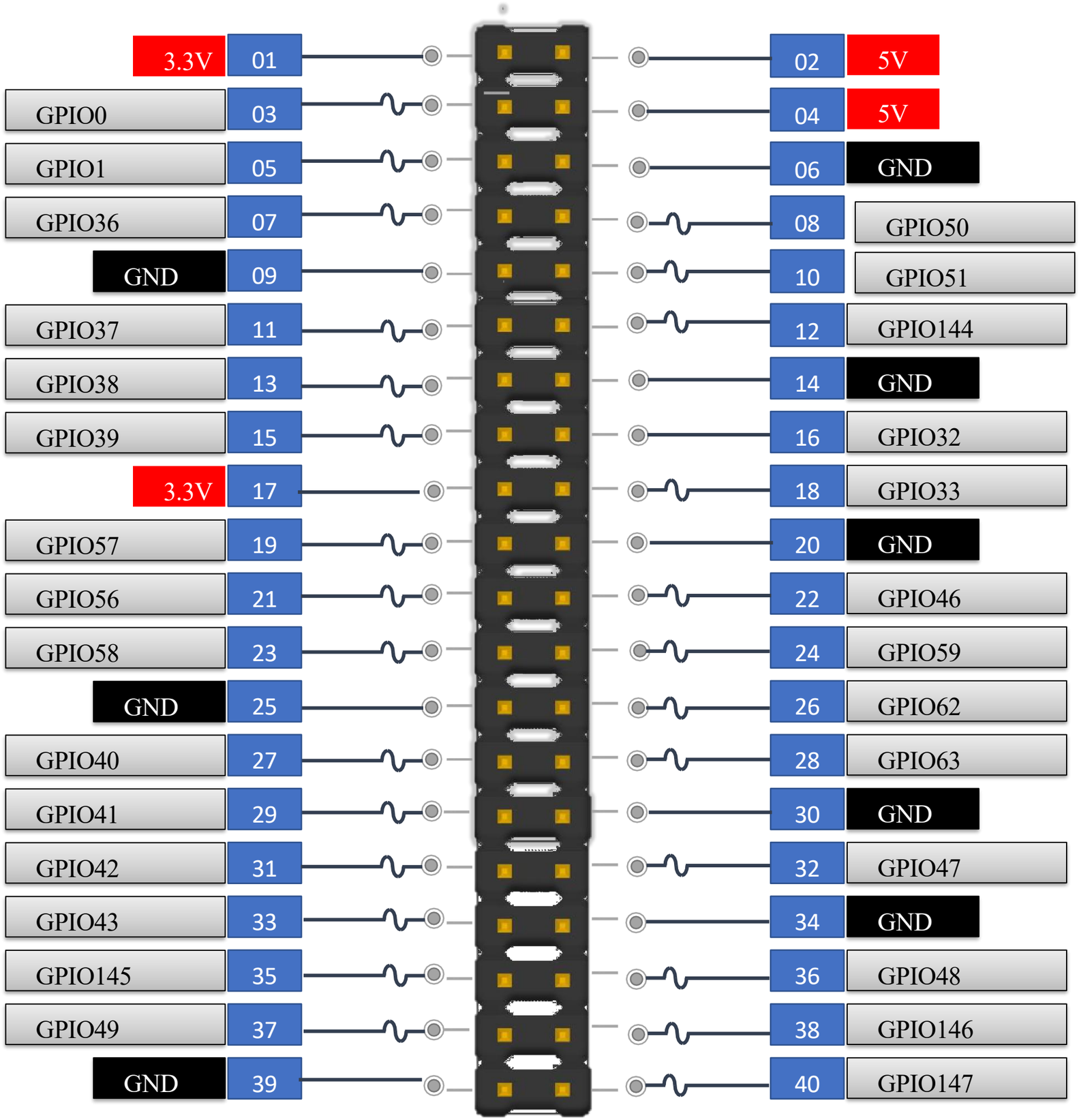
GPIO161		GPIO97	
GPIO162		GPIO98	
GPIO26		GPIO99	
GPIO27		GPIO100	
GPIO163		GPIO101	
GPIO164		GPIO102	
GPIO165		GPIO103	
GPIO166		GPIO104	
NC		GPIO105	
NC		GPIO106	
GPIO34		GPIO107	
GPIO35		GPIO108	
GPIO96		GPIO49	
5V		5V	
GND		3.3V	



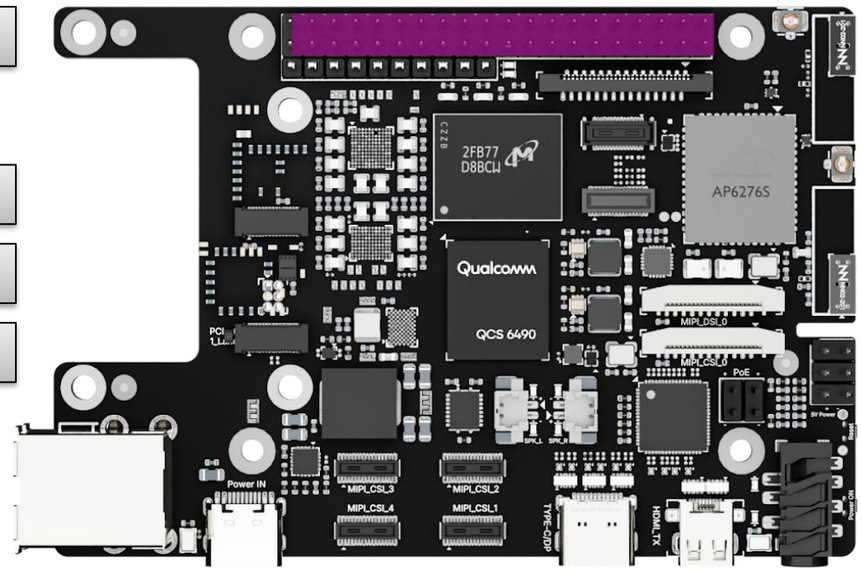
30th



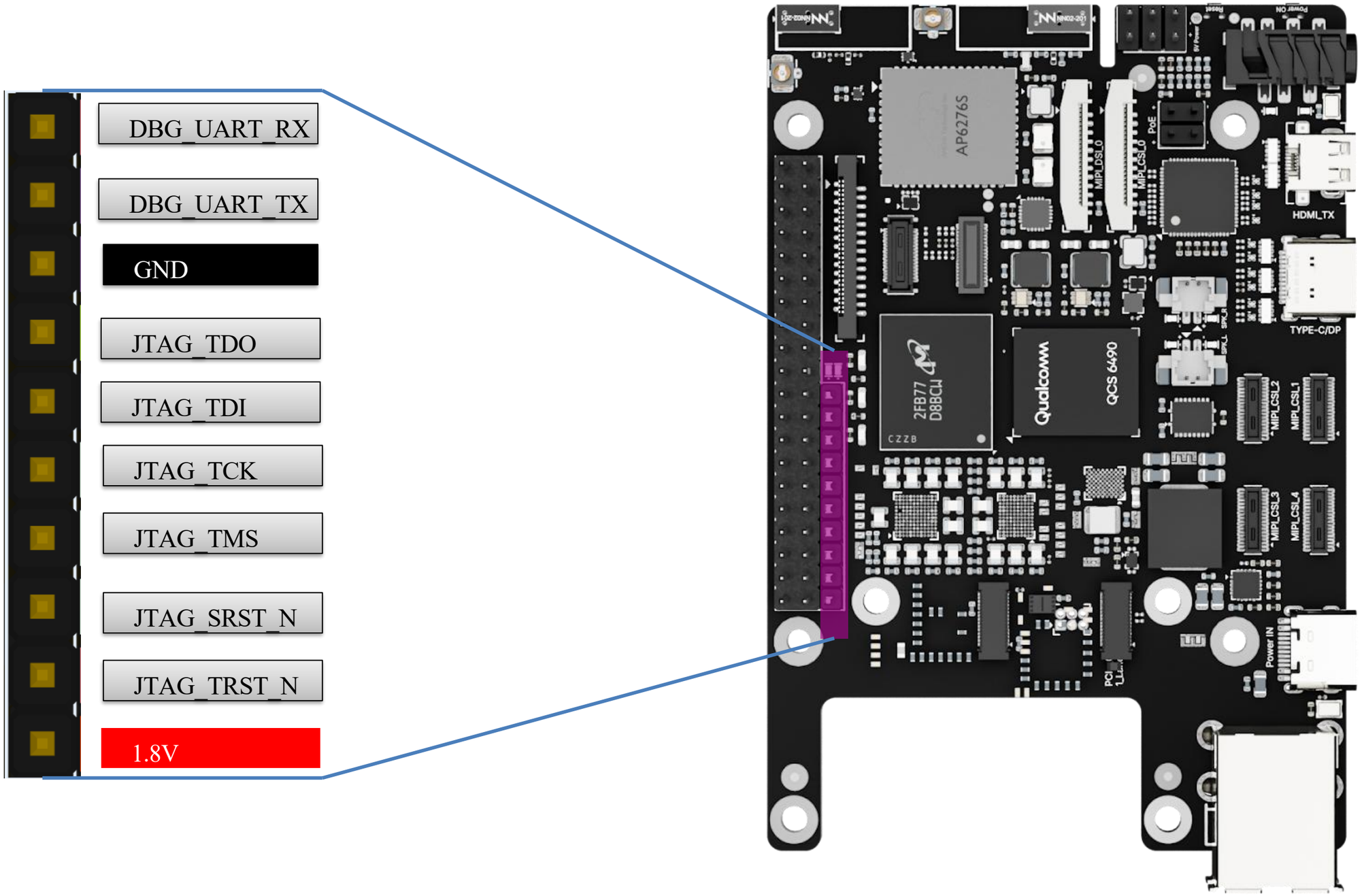
QCS6490 GPIOs



40Pin GPIO Header 2.54mm



Debug Header



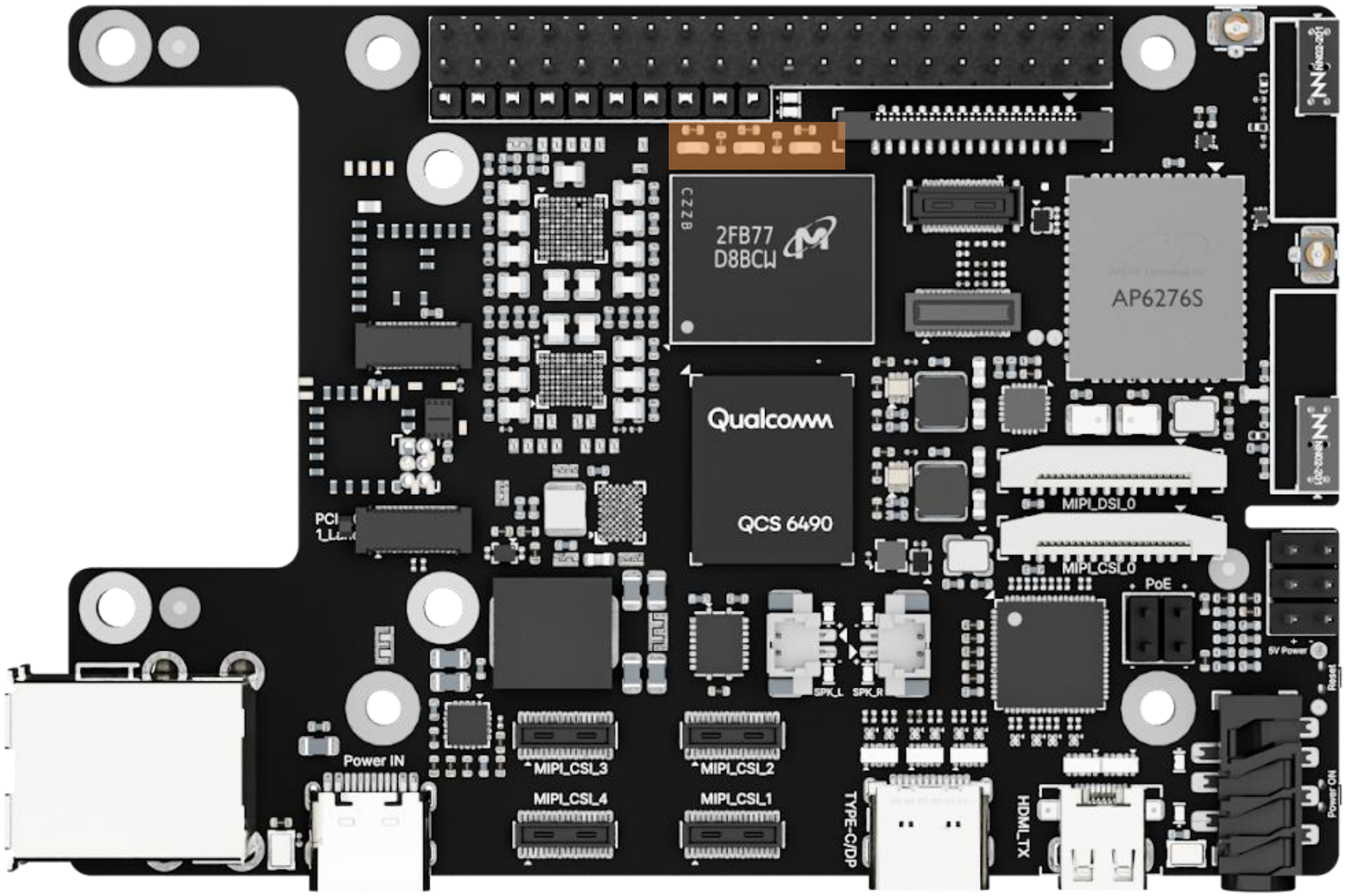
Power, Status & User LED



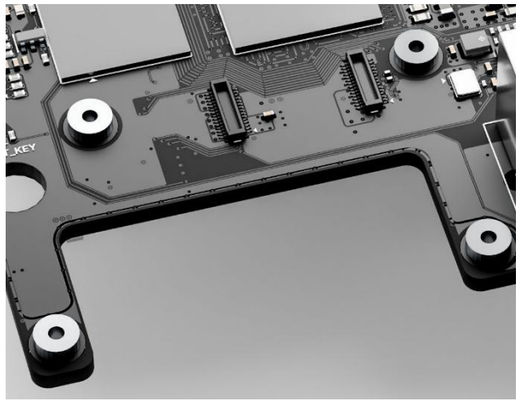
Power LED



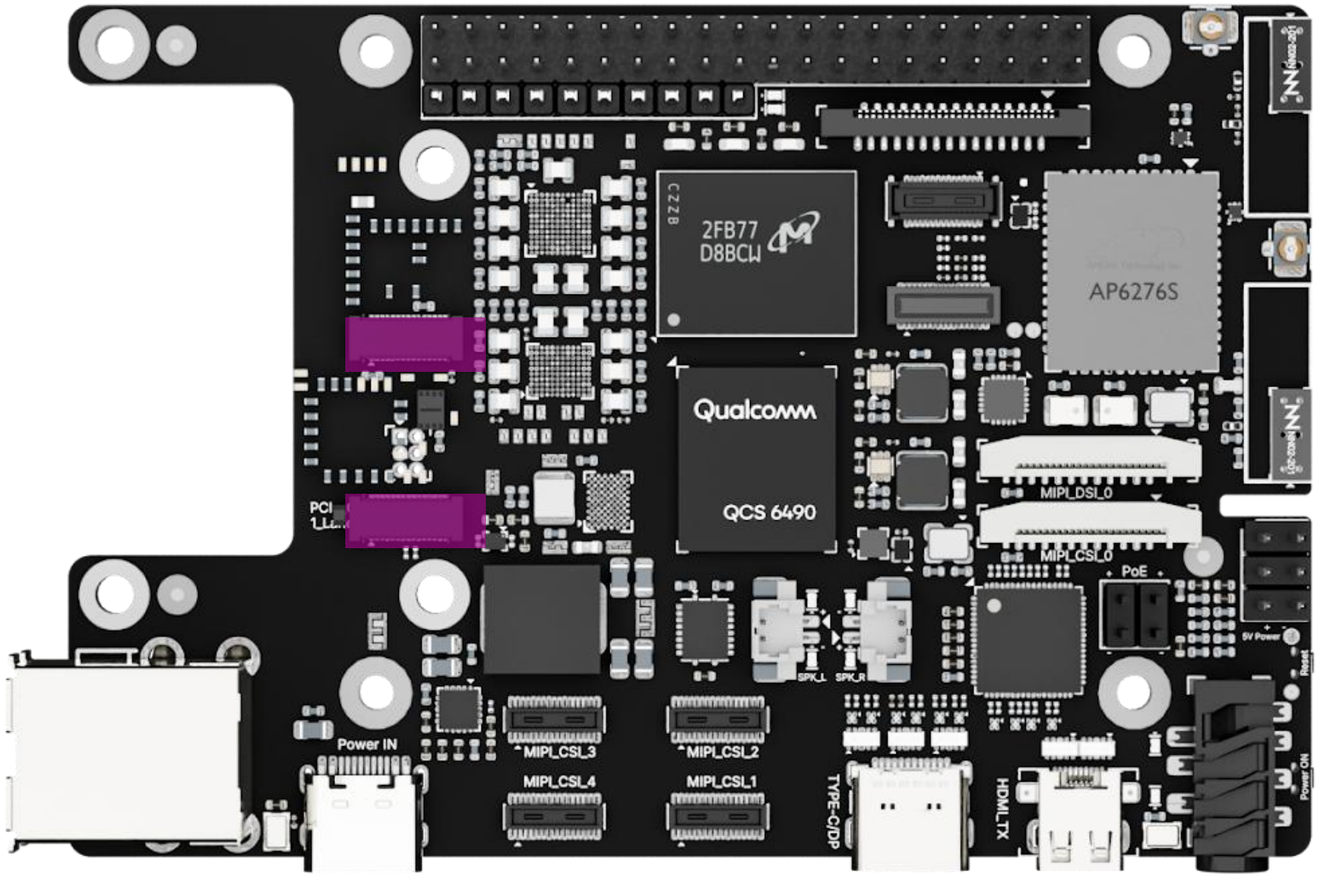
Status & User LED



PCIe Modular Port



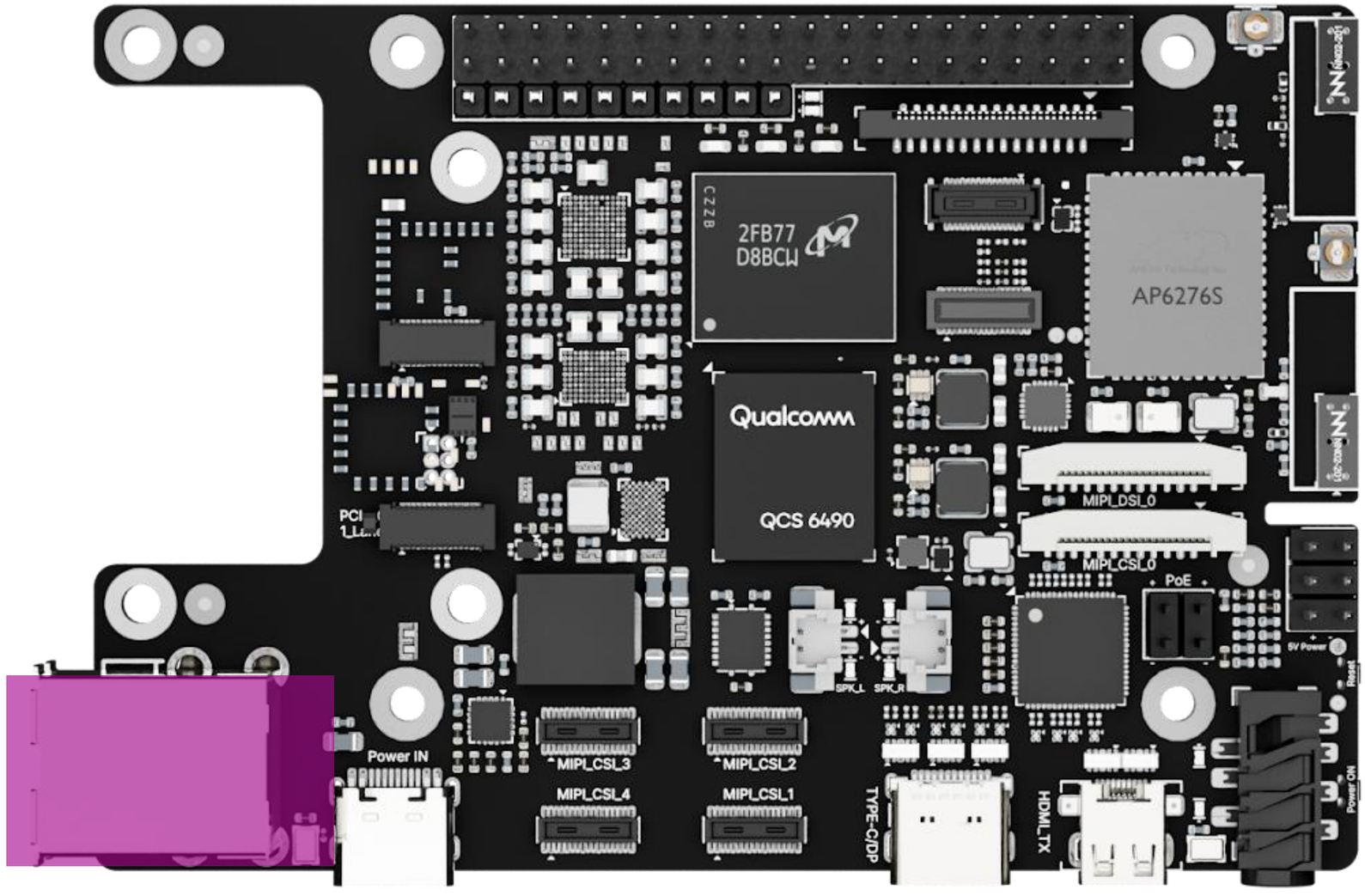
Swappable PCIe Modular Port



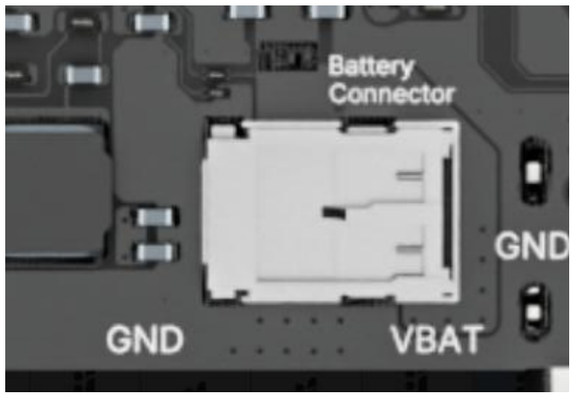
17 2 x USB2.0 Port



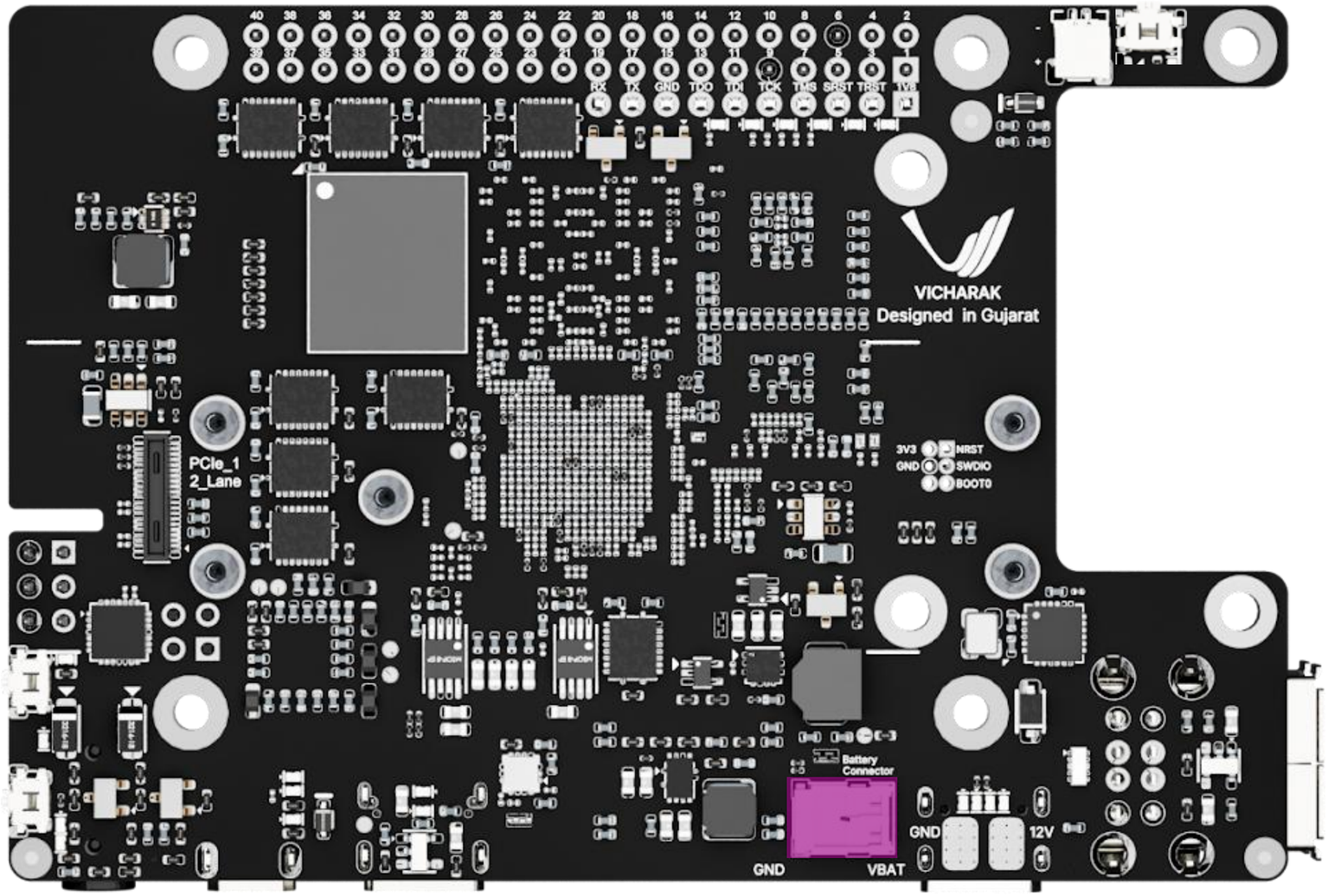
2 x USB 2.0 HOST



Battery Connector

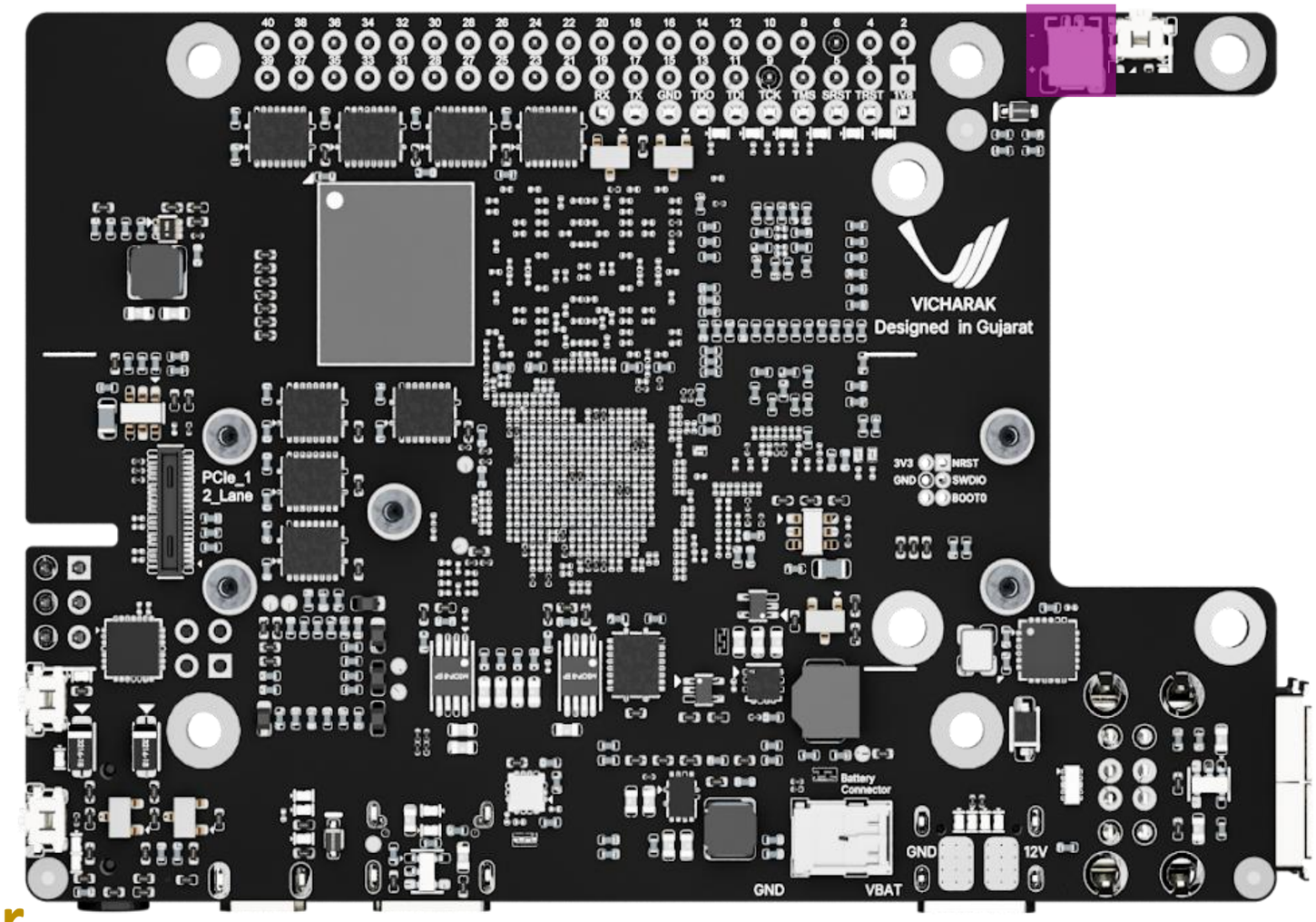


Battery Connector



RTC Battery Connector

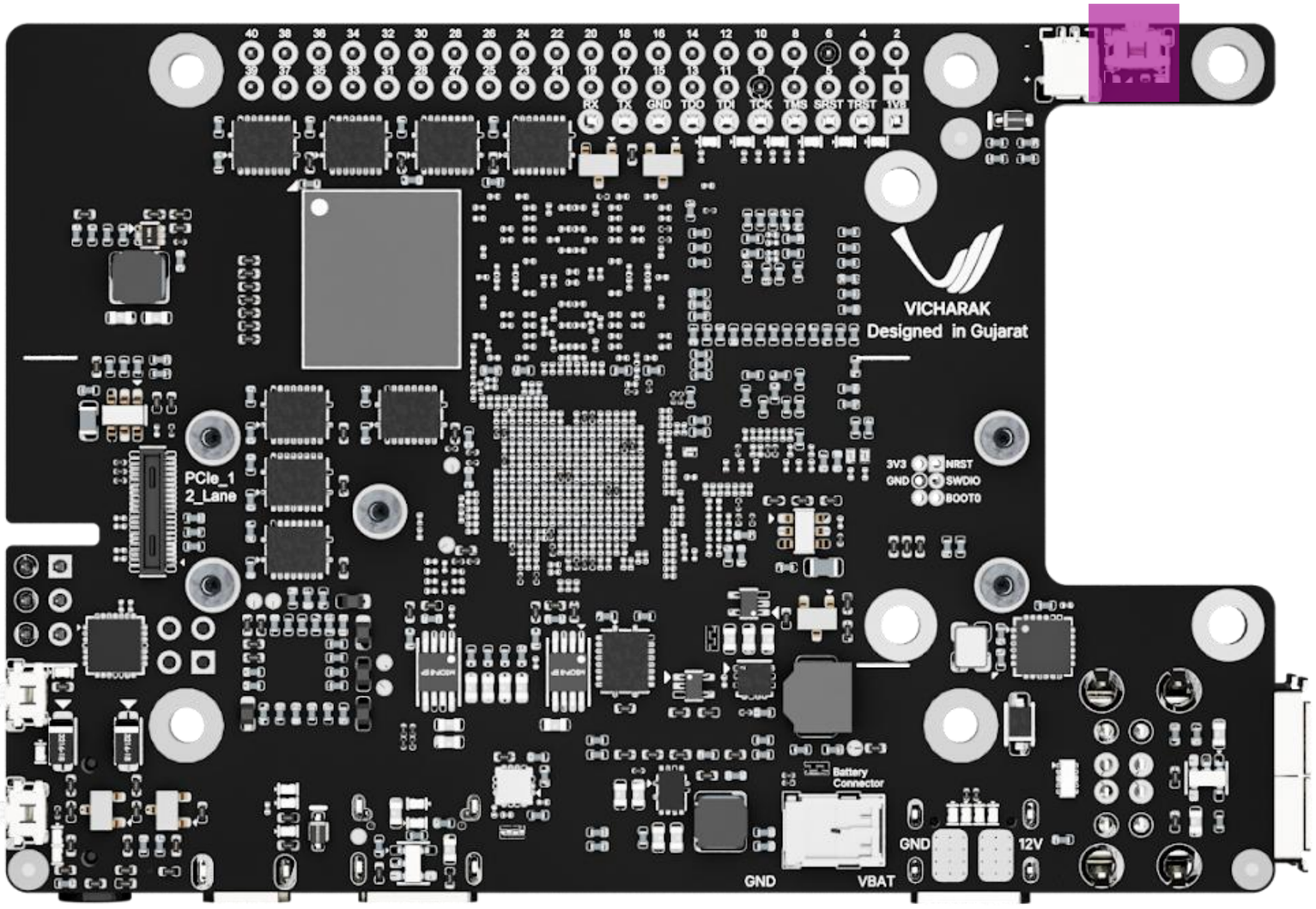




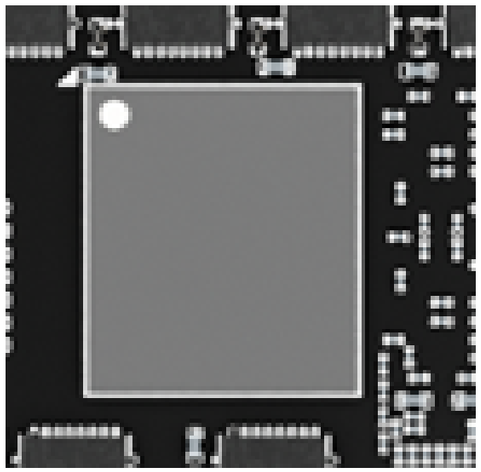
RTC Battery Connector



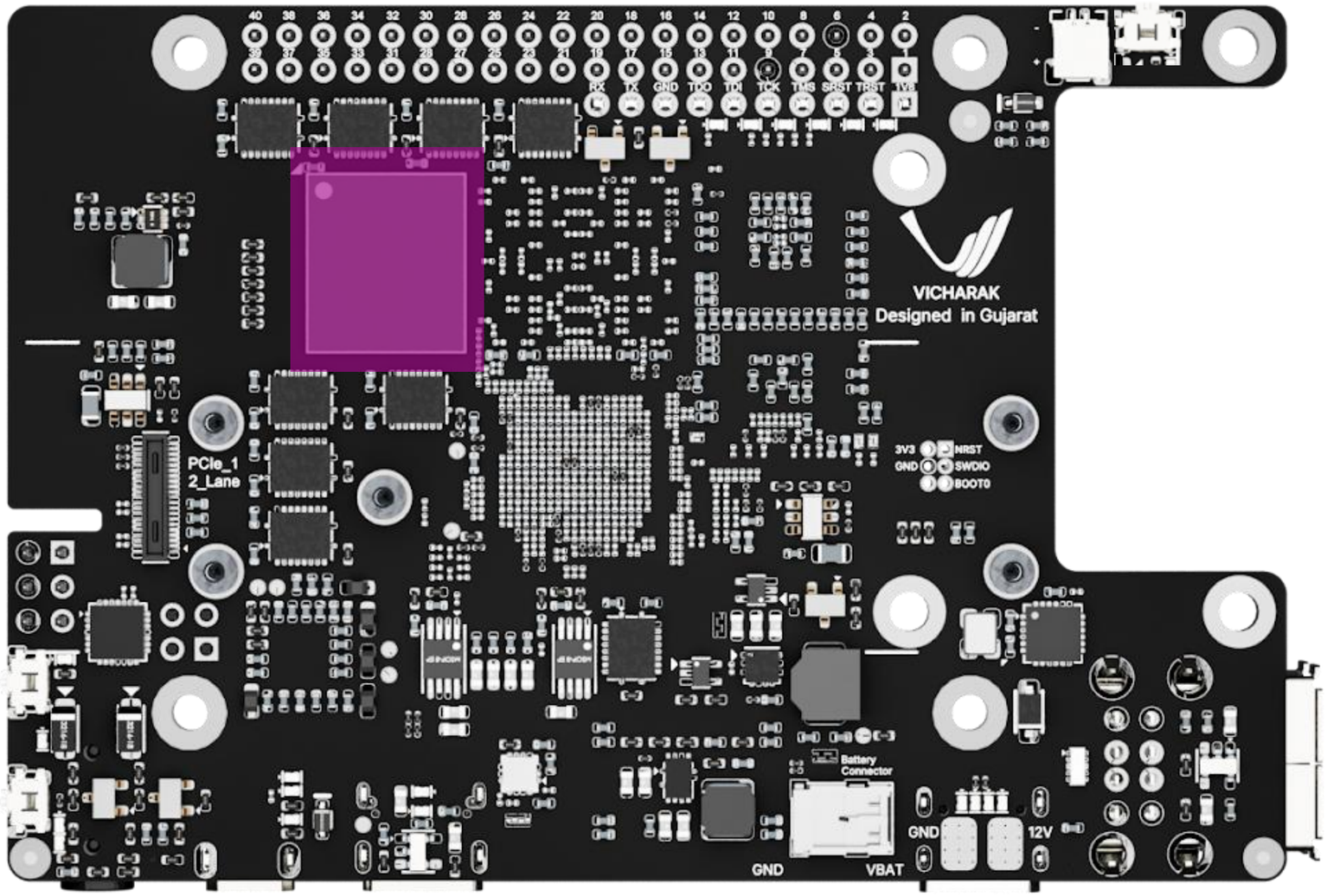
eDL Key

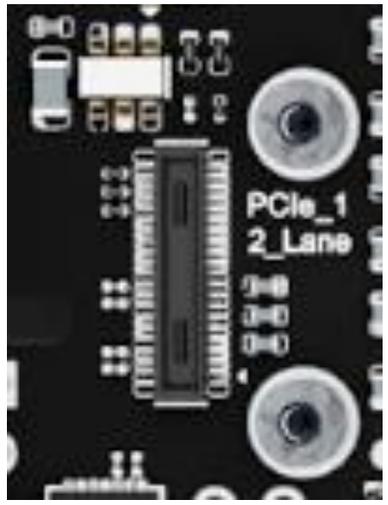


Maskrom Key

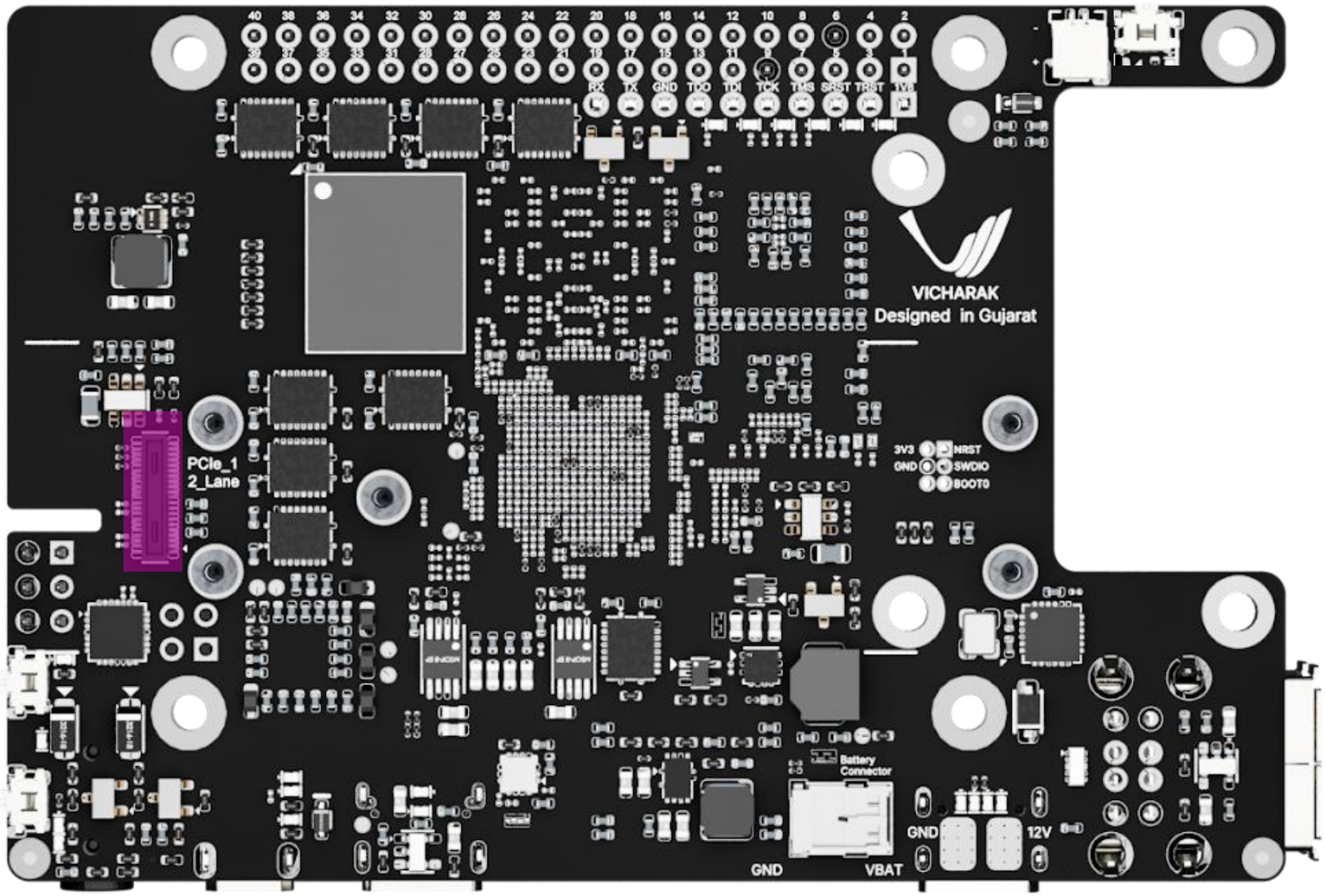


Onboard EMMC Storage

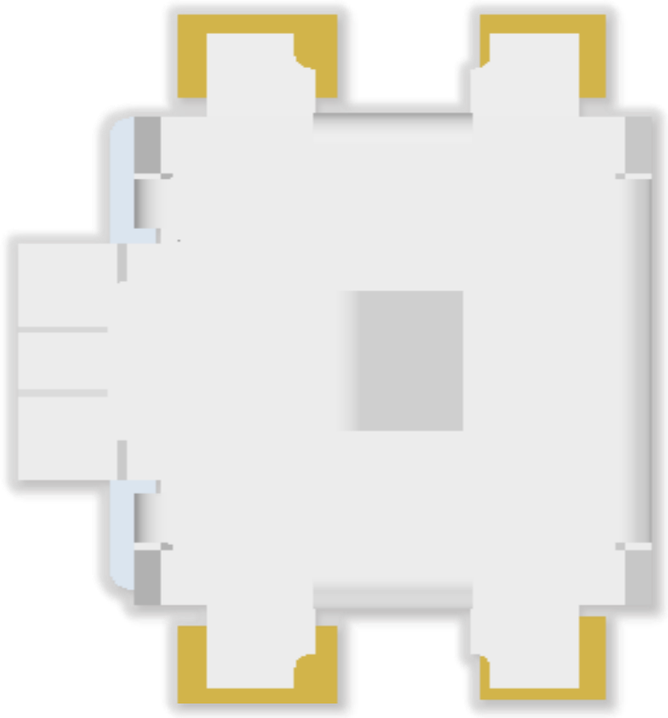




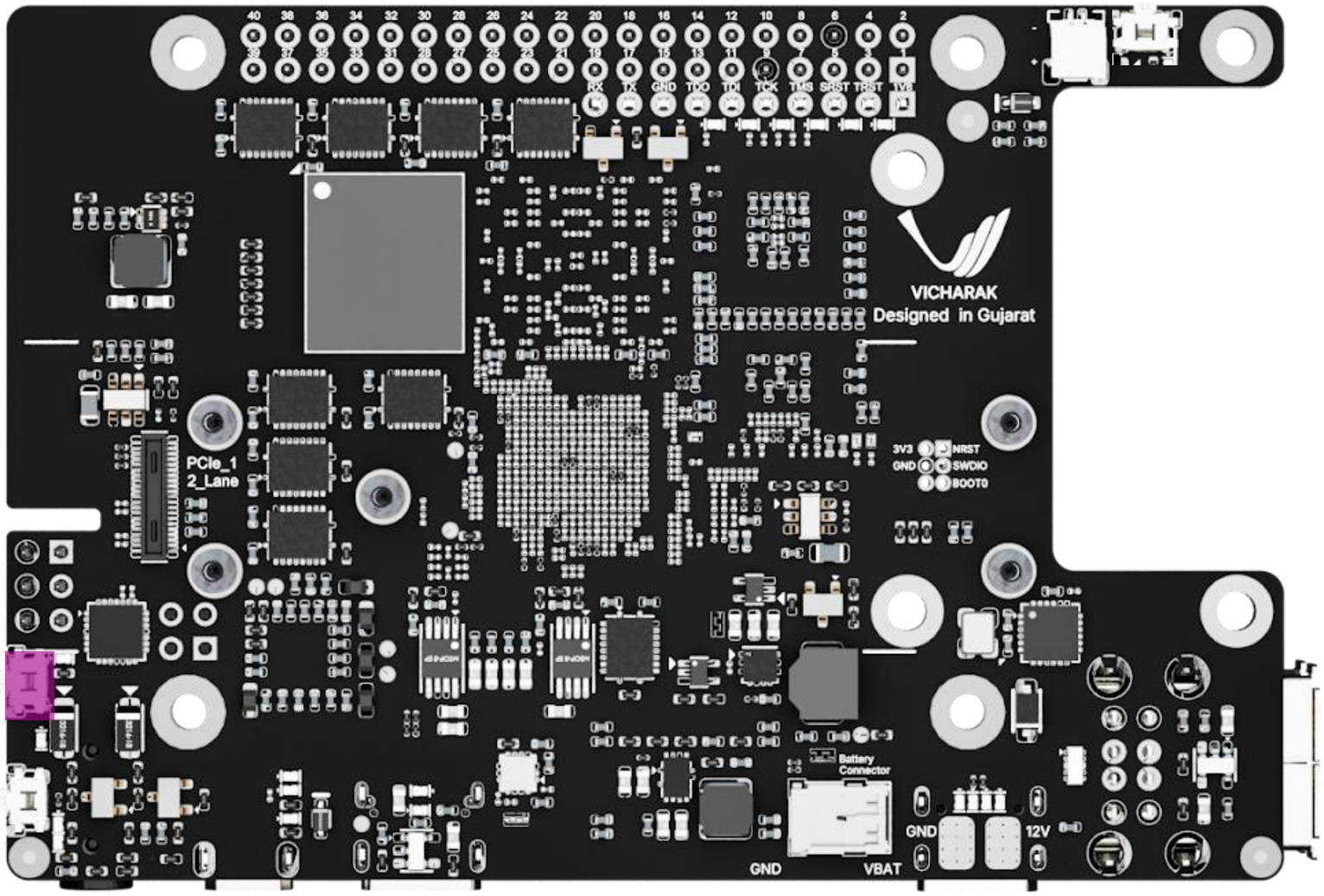
Modular PCIe3 x2 Connector for M key, B key or E key



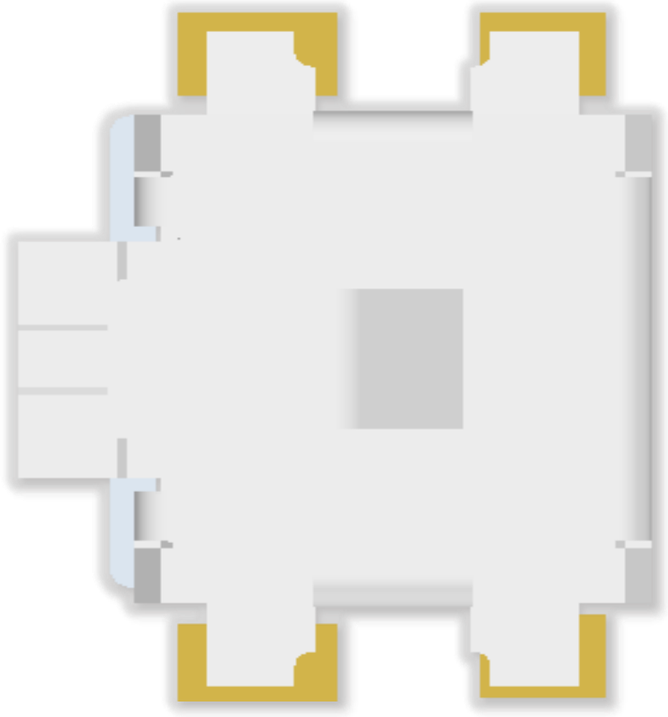
23 Reset key



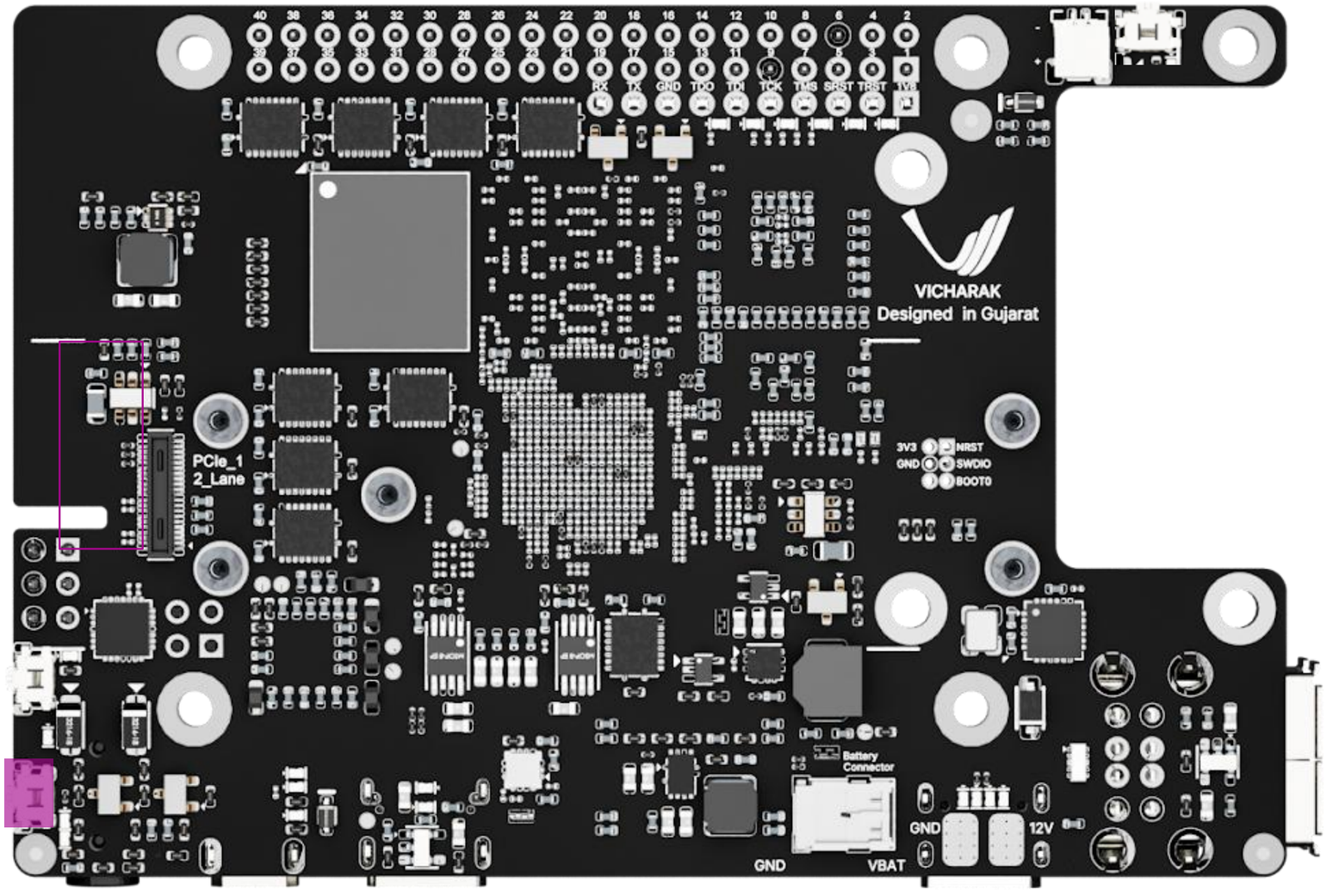
Reset Key



Power key



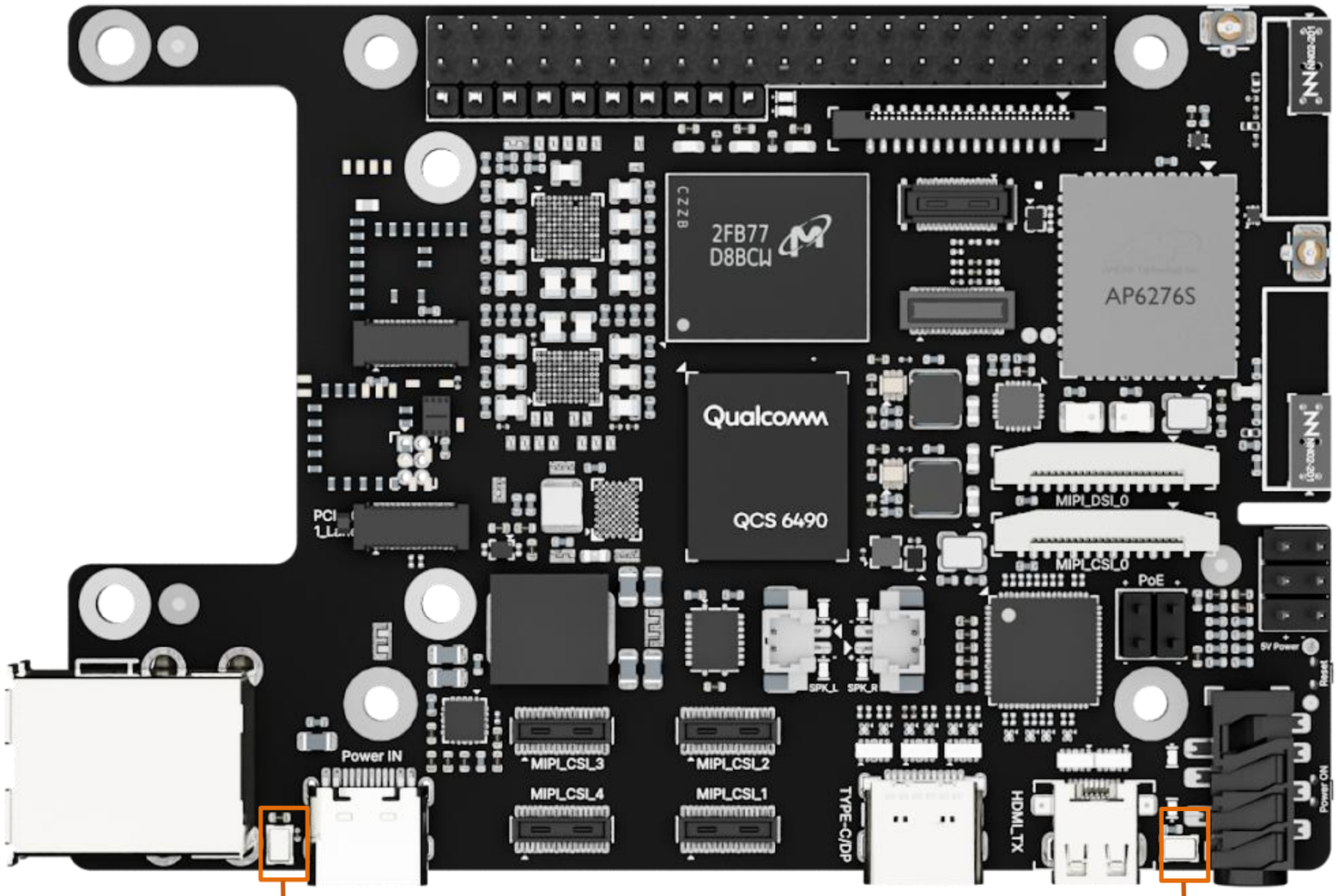
Power Key



MEMS Mics

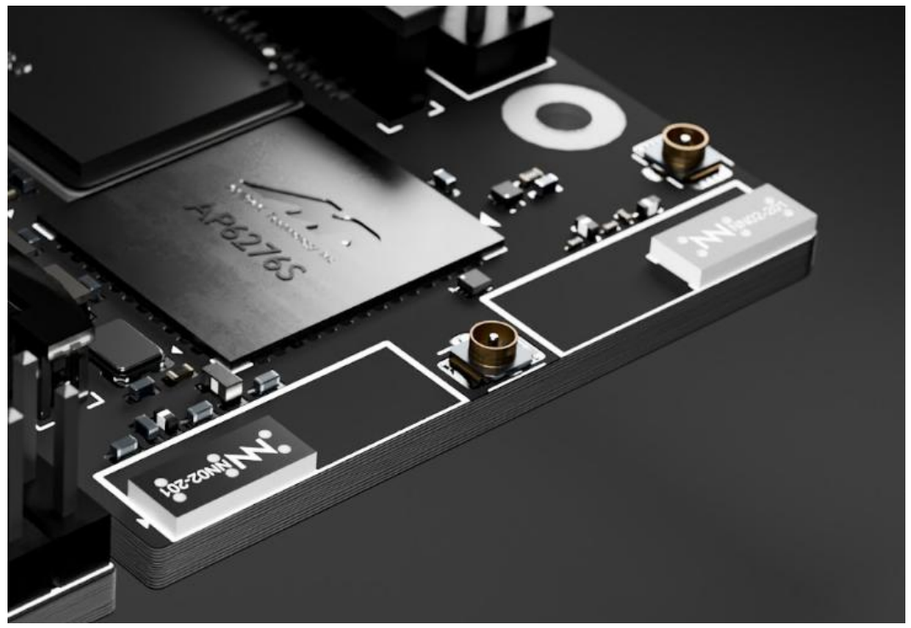


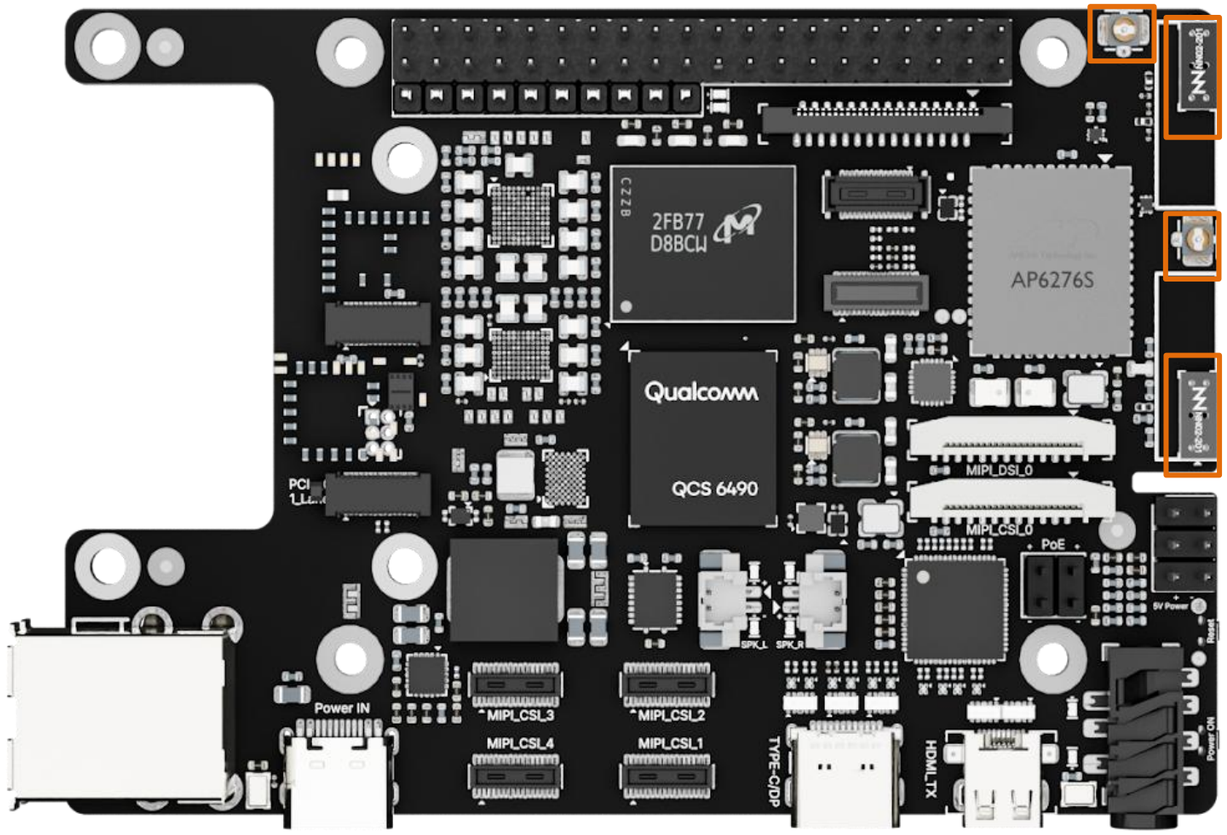
Left & Right MEMS
Digital Mic



Left

Right





U.FL and Chip Antenna Configuration

AXON Mini supports two antenna options: an onboard chip antenna and an external antenna connected through a U.FL connector.

Users can select the desired antenna by configuring the corresponding GPIO, which controls the RF switch to route the RF signal to the selected antenna.